Theory of Algebraic Geometry 1

July 29, 2026

Lecture notes with worked examples in algebraic geometry

Contents

Problem 1 - Solutions: Basic Open Sets, Spectra of Polynomial Rings, and Disconnected Spectra

Let A be a ring. Show that the sets

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\},$$

with f ranging over all elements of A, form a basis of the topology on $\operatorname{Spec} A$. Show that if $D(f) = \varnothing$, then f is nilpotent, i.e. there exists $n > 0$ such that $f^n = 0$.

Solution.

Recall that the Zariski closed subsets of $\operatorname{Spec} A$ are sets of the form

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\},$$

where $I \subseteq A$ is an ideal.

For a single element $f \in A$, we have

$$V((f)) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \in \mathfrak{p}\}.$$

Therefore

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\} = \operatorname{Spec} A \setminus V((f)).$$

Hence every $D(f)$ is open.

We now show that the sets $D(f)$ form a basis.

First,

$$D(1) = \operatorname{Spec} A,$$

because no prime ideal contains 1. Thus the sets $D(f)$ cover $\operatorname{Spec} A$.

Second, for any $f, g \in A$, we have

$$D(f) \cap D(g) = D(fg).$$

Indeed,

$$\mathfrak{p} \in D(f) \cap D(g)$$

if and only if

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$fg \notin \mathfrak{p}.$$

Therefore

$$D(f) \cap D(g) = D(fg).$$

Now let $U \subseteq \operatorname{Spec} A$ be open. Then

$$U = \operatorname{Spec} A \setminus V(I)$$

for some ideal $I \subseteq A$. We claim that

$$U = \bigcup_{f \in I} D(f).$$

Indeed,

$$\mathfrak{p} \in \operatorname{Spec} A \setminus V(I)$$

if and only if

$$I \not\subseteq \mathfrak{p}.$$

This means that there exists $f \in I$ such that

$$f \notin \mathfrak{p}.$$

Equivalently,

$$\mathfrak{p} \in D(f)$$

for some $f \in I$. Hence

$$\operatorname{Spec} A \setminus V(I) = \bigcup_{f \in I} D(f).$$

Therefore every open set is a union of basic opens $D(f)$, so the sets $D(f)$ form a basis for the topology on $\operatorname{Spec} A$.

Now we prove the nilpotence statement.

We first recall the classical identity

$$\sqrt{(0)} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

The ideal $\sqrt{(0)}$ is called the nilradical of A, and it consists exactly of the nilpotent elements of A.

Suppose first that

$$D(f) = \varnothing.$$

Then there is no prime ideal $\mathfrak{p}$ such that $f \notin \mathfrak{p}$. Equivalently,

$$f \in \mathfrak{p}$$

for every prime ideal $\mathfrak{p} \subseteq A$. Hence

$$f \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Using the identity above, we get

$$f \in \sqrt{(0)}.$$

Therefore f is nilpotent. Hence there exists $n > 0$ such that

$$f^n = 0.$$

Conversely, suppose f is nilpotent. Then there exists $n > 0$ such that

$$f^n = 0.$$

Let $\mathfrak{p} \in \operatorname{Spec} A$. Since $0 \in \mathfrak{p}$, we have

$$f^n \in \mathfrak{p}.$$

Because $\mathfrak{p}$ is prime, this implies

$$f \in \mathfrak{p}.$$

Thus every prime ideal contains f. Therefore there is no prime ideal avoiding f, and so

$$D(f) = \varnothing.$$

Thus

$$\boxed{D(f) = \varnothing \iff f \text{ is nilpotent.}}$$

Problem 2

Describe the topological spaces

$$\operatorname{Spec} \mathbb{R}[x], \qquad \operatorname{Spec} \mathbb{C}[x, y], \qquad \operatorname{Spec} \mathbb{Z}[x], \qquad \operatorname{Spec} \mathbb{F}_p[x],$$

where $\mathbb{F}_p$ is the field with p elements.

Solution.

We describe the points and the closure relations in each case.

1. The space $\operatorname{Spec} \mathbb{R}[x]$

The ring $\mathbb{R}[x]$ is a principal ideal domain. Therefore its prime ideals are

$$(0)$$

and the ideals generated by irreducible polynomials over $\mathbb{R}$.

Over $\mathbb{R}$, the irreducible polynomials are of two types:

1) linear polynomials,
$$x - a, \qquad a \in \mathbb{R};$$

2) quadratic polynomials with negative discriminant.

Hence the points of $\operatorname{Spec} \mathbb{R}[x]$ are

$$(0),$$

the ideals

$$(x - a), \qquad a \in \mathbb{R},$$

and the ideals

$$(q(x)),$$

where $q(x) \in \mathbb{R}[x]$ is an irreducible quadratic polynomial.

The point (0) is the generic point of the whole space. Indeed,

$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{R}[x].$$

This happens because

$$V((0)) = \operatorname{Spec} \mathbb{R}[x].$$

Every nonzero prime ideal in $\mathbb{R}[x]$ is maximal, because $\mathbb{R}[x]$ is a PID and any nonzero prime ideal is generated by an irreducible polynomial. Thus all nonzero prime ideals are closed points.

Therefore the closed points are

$$(x-a), \qquad a \in \mathbb{R},$$

and

$$(q(x)),$$

where $q(x)$ is irreducible quadratic over $\mathbb{R}$.

Geometrically, $\operatorname{Spec} \mathbb{R}[x]$ is the affine line over $\mathbb{R}$, but with closed points corresponding not only to real roots, but also to conjugate pairs of complex non-real roots.

The points $(x-a)$ correspond to real points

$$a \in \mathbb{R}.$$

The quadratic ideals correspond to conjugate pairs

$$\alpha, \overline{\alpha} \in \mathbb{C} \setminus \mathbb{R}.$$

Thus $\operatorname{Spec} \mathbb{R}[x]$ is a one-dimensional topological space with one generic point and many closed points.

The proper closed subsets are finite sets of closed points.

2. The space $\operatorname{Spec} \mathbb{C}[x,y]$

The ring $\mathbb{C}[x,y]$ is the coordinate ring of the affine plane over $\mathbb{C}$. Thus $\operatorname{Spec} \mathbb{C}[x,y]$ is the affine plane

$$\mathbb{A}^2_{\mathbb{C}}$$

together with generic points for irreducible algebraic subsets.

The ring $\mathbb{C}[x,y]$ has Krull dimension 2. Therefore its prime ideals have height 0, 1, or 2.

There is one height 0 prime ideal:

$$(0).$$

This is the generic point of the whole affine plane. Its closure is

$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{C}[x,y].$$

The height 1 prime ideals are of the form

$$(f),$$

where $f \in \mathbb{C}[x, y]$ is an irreducible polynomial. These points are not closed. Each such point is the generic point of the irreducible algebraic curve defined by

$$f(x, y) = 0.$$

Its closure is

$$\overline{\{(f)\}} = V(f).$$

The height 2 prime ideals are maximal ideals. By Hilbert's Nullstellensatz, the maximal ideals of $\mathbb{C}[x, y]$ are exactly

$$(x - a, y - b), \qquad (a, b) \in \mathbb{C}^2.$$

These are the closed points.

Therefore the points of $\operatorname{Spec} \mathbb{C}[x, y]$ are:

1) the generic point
$$(0);$$
2) generic points of irreducible curves,
$$(f), \qquad f \in \mathbb{C}[x, y] \text{ irreducible};$$
3) closed points,
$$(x - a, y - b), \qquad (a, b) \in \mathbb{C}^2.$$

The closure relations are:

$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{C}[x, y],$$
$$\overline{\{(f)\}} = V(f),$$

and

$$\overline{\{(x - a, y - b)\}} = \{(x - a, y - b)\}.$$

Thus $\operatorname{Spec} \mathbb{C}[x, y]$ is the affine plane, but topologically it contains more than ordinary complex points. It also contains generic points for curves and a generic point for the whole plane.

3. The space $\operatorname{Spec} \mathbb{Z}[x]$

The ring $\mathbb{Z}[x]$ has dimension 2. It may be viewed as the coordinate ring of the affine line over $\operatorname{Spec} \mathbb{Z}$. Therefore

$$\operatorname{Spec} \mathbb{Z}[x]$$

is an arithmetic surface.

There is one height 0 prime:

$$(0).$$

This is the generic point of the whole space:

$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{Z}[x].$$

The height 1 primes include two important types.

First, for every prime number p, we have the prime ideal

$$(p) \subseteq \mathbb{Z}[x].$$

Indeed,

$$\mathbb{Z}[x]/(p) \cong \mathbb{F}_p[x],$$

which is an integral domain. Therefore (p) is prime.

The closed subset

$$V(p) \subseteq \operatorname{Spec} \mathbb{Z}[x]$$

is the fiber over the prime p. Since

$$\mathbb{Z}[x]/(p) \cong \mathbb{F}_p[x],$$

we have

$$V(p) \cong \operatorname{Spec} \mathbb{F}_p[x].$$

These are called vertical curves.

Second, if $F(x) \in \mathbb{Z}[x]$ is primitive and irreducible, then

$$(F(x))$$

is a height 1 prime ideal. These are called horizontal curves, because they spread over $\operatorname{Spec} \mathbb{Z}$.

The maximal ideals of $\mathbb{Z}[x]$ are the closed points. They are of the form

$$(p, \overline{f}(x)),$$

where p is a prime number and $\overline{f}(x) \in \mathbb{F}_p[x]$ is an irreducible polynomial.

Equivalently, if $f(x) \in \mathbb{Z}[x]$ is a lift of $\overline{f}(x)$, then the maximal ideal may be written as

$$(p, f(x)).$$

The residue field is

$$\mathbb{Z}[x]/(p, f(x)) \cong \mathbb{F}_p[x]/(\overline{f}(x)),$$

which is a finite field.

Therefore the points of $\operatorname{Spec} \mathbb{Z}[x]$ are:

1) the generic point

$$(0);$$

2) vertical curve points
$$(p),$$
where p is a prime number;

3) horizontal curve points
$$(F(x)),$$
where $F(x) \in \mathbb{Z}[x]$ is primitive and irreducible;

4) closed points
$$(p, \overline{f}(x)),$$
where p is prime and $\overline{f}(x) \in \mathbb{F}_p[x]$ is irreducible.

The closure relations are as follows.

The closure of the generic point is the whole space:
$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{Z}[x].$$

The closure of a height 1 point is an irreducible curve:
$$\overline{\{(p)\}} = V(p) \cong \operatorname{Spec} \mathbb{F}_p[x],$$
and
$$\overline{\{(F)\}} = V(F).$$

The maximal ideals are closed points:
$$\overline{\{(p, \overline{f})\}} = \{(p, \overline{f})\}.$$

Thus $\operatorname{Spec} \mathbb{Z}[x]$ is a two-dimensional arithmetic space. It has a generic point, one-dimensional curve points, and closed points with finite residue fields.

4. The space $\operatorname{Spec} \mathbb{F}_p[x]$

The ring $\mathbb{F}_p[x]$ is a principal ideal domain. Therefore its prime ideals are
$$(0)$$
and the ideals generated by irreducible polynomials.

Hence the points of $\operatorname{Spec} \mathbb{F}_p[x]$ are
$$(0)$$
and
$$(f(x)),$$
where $f(x) \in \mathbb{F}_p[x]$ is irreducible.

The point (0) is the generic point:
$$\overline{\{(0)\}} = \operatorname{Spec} \mathbb{F}_p[x].$$

Every nonzero prime ideal in $\mathbb{F}_p[x]$ is maximal. Therefore each

$$(f(x))$$

with f irreducible is a closed point.

The linear ideals

$$(x-a), \qquad a \in \mathbb{F}_p,$$

correspond to the p rational points of the affine line over $\mathbb{F}_p$.

If $f(x)$ is irreducible of degree $d > 1$, then the residue field is

$$\mathbb{F}_p[x]/(f(x)) \cong \mathbb{F}_{p^d}.$$

Thus irreducible polynomials of higher degree correspond to closed points whose residue fields are finite extensions of $\mathbb{F}_p$.

Therefore $\operatorname{Spec} \mathbb{F}_p[x]$ is the affine line over $\mathbb{F}_p$, but its closed points are not only the p rational points. They also include points corresponding to irreducible polynomials of all degrees.

As in the case of a polynomial ring in one variable over a field, the proper closed subsets are finite sets of closed points.

Problem 3

Let A be a ring. Show that the following are equivalent:

i) $\operatorname{Spec} A$ is disconnected.

ii) There exist nonzero elements $e_1, e_2 \in A$ such that

$$e_1 e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

iii) A is isomorphic to $A_1 \times A_2$ for some nonzero rings A_1, A_2.

Solution.

The idea is that disconnectedness of $\operatorname{Spec} A$ is the same thing as the existence of a nontrivial idempotent decomposition of the ring.

An element $e \in A$ is called idempotent if

$$e^2 = e.$$

The trivial idempotents are 0 and 1. A nontrivial idempotent is an idempotent different from 0 and 1.

We prove the implications one by one.

$(ii) \Rightarrow (i)$

Assume that there exist nonzero elements $e_1, e_2 \in A$ such that

$$e_1e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

Consider the two basic open sets

$$D(e_1) \qquad \text{and} \qquad D(e_2).$$

Their intersection is

$$D(e_1) \cap D(e_2) = D(e_1e_2).$$

Since $e_1e_2 = 0$, this becomes

$$D(e_1) \cap D(e_2) = D(0).$$

But

$$D(0) = \{\mathfrak{p} \in \operatorname{Spec} A \mid 0 \notin \mathfrak{p}\} = \varnothing,$$

because every ideal contains 0. Hence

$$D(e_1) \cap D(e_2) = \varnothing.$$

Now we show that

$$D(e_1) \cup D(e_2) = \operatorname{Spec} A.$$

Let $\mathfrak{p} \in \operatorname{Spec} A$. Suppose, for contradiction, that

$$\mathfrak{p} \notin D(e_1) \cup D(e_2).$$

Then

$$e_1 \in \mathfrak{p} \quad \text{and} \quad e_2 \in \mathfrak{p}.$$

Therefore

$$e_1 + e_2 \in \mathfrak{p}.$$

But

$$e_1 + e_2 = 1.$$

Thus

$$1 \in \mathfrak{p},$$

which is impossible because a prime ideal is proper. Hence

$$D(e_1) \cup D(e_2) = \operatorname{Spec} A.$$

It remains to see that both $D(e_1)$ and $D(e_2)$ are nonempty.

If $D(e_1) = \varnothing$, then by Problem 1, e_1 is nilpotent. But e_1 is idempotent:

$$e_1^2 = e_1.$$

If an idempotent is nilpotent, then it must be zero. Indeed, if $e_1^n = 0$, then for every $n \geq 1$,

$$e_1^n = e_1.$$

Thus $e_1 = 0$, contradicting the assumption that $e_1 \neq 0$. Therefore

$$D(e_1) \neq \varnothing.$$

Similarly,

$$D(e_2) \neq \varnothing.$$

Therefore

$$\operatorname{Spec} A = D(e_1) \sqcup D(e_2)$$

is a decomposition of $\operatorname{Spec} A$ into two disjoint nonempty open subsets. Hence $\operatorname{Spec} A$ is disconnected.

$(ii) \Rightarrow (iii)$

Assume again that

$$e_1 e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

Define

$$A_1 = Ae_1, \qquad A_2 = Ae_2.$$

These are ideals of A, but they are also rings in their own right. The identity element of Ae_1 is e_1, and the identity element of Ae_2 is e_2.

Define a map

$$\varphi : A \longrightarrow Ae_1 \times Ae_2$$

by

$$\varphi(a) = (ae_1, ae_2).$$

We show that φ is a ring isomorphism.

First, φ is additive:

$$\varphi(a+b) = ((a+b)e_1, (a+b)e_2) = (ae_1 + be_1, ae_2 + be_2) = \varphi(a) + \varphi(b).$$

It is multiplicative:

$$\varphi(ab) = (abe_1, abe_2).$$

On the other hand,

$$\varphi(a)\varphi(b) = (ae_1, ae_2)(be_1, be_2) = (ae_1 be_1, ae_2 be_2).$$

Since $e_1^2 = e_1$ and $e_2^2 = e_2$, we get

$$ae_1 be_1 = abe_1^2 = abe_1,$$

and

$$ae_2 be_2 = abe_2^2 = abe_2.$$

Therefore
$$\varphi(a)\varphi(b) = \varphi(ab).$$

Also,
$$\varphi(1) = (e_1, e_2),$$
which is the identity element of $Ae_1 \times Ae_2$.

Now define
$$\psi : Ae_1 \times Ae_2 \longrightarrow A$$
by
$$\psi(x, y) = x + y.$$

Let $a \in A$. Then
$$\psi(\varphi(a)) = \psi(ae_1, ae_2) = ae_1 + ae_2 = a(e_1 + e_2) = a.$$

Thus
$$\psi \circ \varphi = \mathrm{id}_A .$$

Conversely, let $x \in Ae_1$ and $y \in Ae_2$. Then there exist $a, b \in A$ such that
$$x = ae_1, \qquad y = be_2.$$

We have
$$xe_1 = x, \qquad xe_2 = 0,$$
and
$$ye_1 = 0, \qquad ye_2 = y.$$

Therefore
$$\varphi(\psi(x, y)) = \varphi(x + y) = ((x + y)e_1, (x + y)e_2) = (xe_1 + ye_1, xe_2 + ye_2) = (x, y).$$

Thus
$$\varphi \circ \psi = \mathrm{id}_{Ae_1 \times Ae_2} .$$

Hence φ is an isomorphism, and
$$A \cong Ae_1 \times Ae_2.$$

Since $e_1, e_2 \neq 0$, both rings Ae_1 and Ae_2 are nonzero.

Therefore condition (iii) holds.

$(iii) \Rightarrow (ii)$

Assume that
$$A \cong A_1 \times A_2,$$
where A_1 and A_2 are nonzero rings.

Inside $A_1 \times A_2$, define
$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$e_1^2 = (1,0)^2 = (1,0) = e_1,$$

and

$$e_2^2 = (0,1)^2 = (0,1) = e_2.$$

Also,

$$e_1e_2 = (1,0)(0,1) = (0,0),$$

and

$$e_1 + e_2 = (1,0) + (0,1) = (1,1),$$

which is the identity element of $A_1 \times A_2$.

Because A_1 and A_2 are nonzero, we have

$$e_1 \neq 0, \qquad e_2 \neq 0.$$

Transporting these elements through the isomorphism

$$A \cong A_1 \times A_2,$$

we obtain nonzero elements $e_1, e_2 \in A$ satisfying

$$e_1e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

Therefore condition (ii) holds.

$(i) \Rightarrow (ii)$

Assume that $\operatorname{Spec} A$ is disconnected. Then there exist nonempty disjoint open subsets $U, V \subseteq \operatorname{Spec} A$ such that

$$\operatorname{Spec} A = U \sqcup V.$$

Since $V = \operatorname{Spec} A \setminus U$, the set U is also closed. Similarly, V is also closed. Hence U and V are both open and closed, that is, clopen.

We use the following standard fact.

Lemma. For a ring A, clopen subsets of $\operatorname{Spec} A$ correspond exactly to idempotents of A. More precisely, if $e \in A$ is idempotent, then

$$D(e)$$

is clopen. Conversely, every clopen subset of $\operatorname{Spec} A$ is of the form

$$D(e)$$

for some idempotent $e \in A$.

Let us first check the easy direction of the lemma. Suppose $e^2 = e$. Then

$$e(1-e) = 0.$$

For a prime ideal $\mathfrak{p}$, since

$$e(1-e) \in \mathfrak{p},$$

we must have

$$e \in \mathfrak{p} \quad \text{or} \quad 1-e \in \mathfrak{p}.$$

Moreover, both cannot occur, because then

$$1 = e + (1-e) \in \mathfrak{p},$$

which is impossible.

Thus, for every prime ideal $\mathfrak{p}$, exactly one of e and $1-e$ belongs to $\mathfrak{p}$. Hence

$$D(e) = V(1-e).$$

Indeed,

$$\mathfrak{p} \in D(e)$$

means

$$e \notin \mathfrak{p},$$

which is equivalent to

$$1-e \in \mathfrak{p}.$$

Therefore

$$D(e) = V(1-e).$$

Thus $D(e)$ is open by definition and closed because it is equal to $V(1-e)$. Hence $D(e)$ is clopen.

Conversely, every clopen subset arises in this way from an idempotent. This is the standard idempotent–clopen correspondence for affine schemes.

Since U is clopen, there exists an idempotent $e_1 \in A$ such that

$$U = D(e_1).$$

Define

$$e_2 = 1 - e_1.$$

Then

$$e_2^2 = (1-e_1)^2 = 1 - 2e_1 + e_1^2.$$

Since $e_1^2 = e_1$, this becomes

$$e_2^2 = 1 - 2e_1 + e_1 = 1 - e_1 = e_2.$$

Therefore e_2 is also idempotent.

Moreover,

$$e_1 e_2 = e_1(1-e_1) = e_1 - e_1^2 = 0,$$

and

$$e_1 + e_2 = e_1 + (1-e_1) = 1.$$

It remains to check that e_1 and e_2 are nonzero.

Because $U \neq \varnothing$, we have

$$D(e_1) \neq \varnothing.$$

If $e_1 = 0$, then

$$D(e_1) = D(0) = \varnothing,$$

contradicting $U \neq \varnothing$. Hence

$$e_1 \neq 0.$$

Similarly, the complement of $D(e_1)$ is

$$V(e_1).$$

Since e_1 is idempotent, this complement is also

$$D(1 - e_1) = D(e_2).$$

But this complement is V, and $V \neq \varnothing$. Therefore

$$D(e_2) \neq \varnothing.$$

Hence

$$e_2 \neq 0.$$

Thus we have found nonzero elements $e_1, e_2 \in A$ such that

$$e_1 e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

Therefore condition (ii) holds.

Conclusion

We have proved

$$(ii) \Rightarrow (i), \qquad (ii) \Rightarrow (iii), \qquad (iii) \Rightarrow (ii), \qquad (i) \Rightarrow (ii).$$

Therefore the three conditions are equivalent:

$$\boxed{\operatorname{Spec} A \text{ is disconnected} \iff A \text{ has a nontrivial idempotent decomposition} \iff A \cong A_1 \times A_2}$$

for some nonzero rings A_1, A_2.

1 The General Picture

The three problems are centered around three fundamental correspondences in affine algebraic geometry:

$$\text{elements } f \in A \quad \longleftrightarrow \quad \text{basic open sets } D(f) \subseteq \operatorname{Spec} A,$$

$$\text{prime ideals of } A \quad \longleftrightarrow \quad \text{points of } \operatorname{Spec} A,$$

and

$$\text{idempotents in } A \quad \longleftrightarrow \quad \text{clopen decompositions of } \operatorname{Spec} A.$$

The main conceptual dictionary is:

$$\boxed{D(f) \text{ is the region where } f \text{ is invertible/nonzero.}}$$

$$\boxed{\operatorname{Spec} A \text{ contains closed points and generic points.}}$$

$$\boxed{\operatorname{Spec} A \text{ is disconnected exactly when } A \text{ splits by idempotents.}}$$

Throughout, A is a commutative ring with identity.

2 Basic Opens $D(f)$

2.1 Definition of the Zariski Topology

Definition 2.1. *Let A be a ring. The spectrum of A, denoted*

$$\operatorname{Spec} A,$$

is the set of all prime ideals of A.

For an ideal $I \subseteq A$, define

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\}.$$

The sets $V(I)$ are the closed sets of the Zariski topology on $\operatorname{Spec} A$.

Thus an ideal I determines the set of prime ideals where all elements of I vanish.

Definition 2.2. *For an element $f \in A$, define*

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}.$$

This is called a basic open subset of $\operatorname{Spec} A$.

Since

$$V((f)) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \in \mathfrak{p}\},$$

we have

$$D(f) = \operatorname{Spec} A \setminus V((f)).$$

Therefore $D(f)$ is open.

2.2 Geometric Meaning of $D(f)$

If A is a coordinate ring, we may think of elements $f \in A$ as functions. Then

$$D(f)$$

is the region where f does not vanish.

For example, if

$$A = k[x, y],$$

then

$$D(x)$$

is the affine plane with the y-axis removed.

Indeed, the condition $x \notin \mathfrak{p}$ means that we are not on the closed set defined by $x = 0$. Therefore

$$D(x) = \mathbb{A}^2_k \setminus V(x).$$

Similarly,

$$D(xy) = D(x) \cap D(y).$$

Geometrically this is the affine plane with both coordinate axes removed.

2.3 Basic Opens Form a Basis

Proposition 2.1. *The sets*

$$D(f), \qquad f \in A,$$

form a basis for the Zariski topology on $\operatorname{Spec} A$.

Proof. First,

$$D(1) = \operatorname{Spec} A,$$

because no prime ideal contains 1. Hence the sets $D(f)$ cover $\operatorname{Spec} A$.

Second, for $f, g \in A$, we have

$$D(f) \cap D(g) = D(fg).$$

Indeed,

$$\mathfrak{p} \in D(f) \cap D(g)$$

if and only if

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$fg \notin \mathfrak{p}.$$

Therefore

$$D(f) \cap D(g) = D(fg).$$

Finally, every open set is a union of such sets. Let $U \subseteq \operatorname{Spec} A$ be open. Then

$$U = \operatorname{Spec} A \setminus V(I)$$

for some ideal $I \subseteq A$. Now

$$\mathfrak{p} \in \operatorname{Spec} A \setminus V(I)$$

if and only if

$$I \not\subseteq \mathfrak{p}.$$

This is equivalent to saying that there exists $f \in I$ such that

$$f \notin \mathfrak{p}.$$

Hence

$$\mathfrak{p} \in D(f)$$

for some $f \in I$. Therefore

$$\operatorname{Spec} A \setminus V(I) = \bigcup_{f \in I} D(f).$$

Thus the sets $D(f)$ form a basis. □

2.4 The Localization Interpretation

The most important algebraic meaning of $D(f)$ is localization.

Theorem 2.1. *For every $f \in A$, there is a natural homeomorphism*

$$D(f) \cong \operatorname{Spec} A_f,$$

where

$$A_f = A\left[\frac{1}{f}\right].$$

Proof. Prime ideals of A_f correspond to prime ideals $\mathfrak{p} \subseteq A$ that do not meet the multiplicative set

$$S = \{1, f, f^2, f^3, \dots\}.$$

This means precisely that

$$f \notin \mathfrak{p}.$$

Therefore prime ideals of A_f correspond exactly to prime ideals of A lying in $D(f)$.

The correspondence is

$$\mathfrak{p} \in D(f) \quad \longmapsto \quad \mathfrak{p}A_f \in \operatorname{Spec} A_f,$$

with inverse

$$\mathfrak{q} \in \operatorname{Spec} A_f \quad \longmapsto \quad \mathfrak{q} \cap A.$$

This gives a homeomorphism

$$D(f) \cong \operatorname{Spec} A_f.$$

□

So $D(f)$ is the part of $\operatorname{Spec} A$ where f is allowed to become invertible.

$$\boxed{D(f) \text{ means: localize } A \text{ by inverting } f.}$$

2.5 Nilpotents and Empty Basic Opens

Definition 2.3. *The nilradical of A is*

$$\mathrm{Nil}(A) = \sqrt{(0)} = \{a \in A \mid a^n = 0 \textit{ for some } n > 0\}.$$

Theorem 2.2. *For every ring A,*

$$\mathrm{Nil}(A) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Proof. If $a \in \mathrm{Nil}(A)$, then $a^n = 0$ for some $n > 0$. Since every prime ideal contains 0, we have

$$a^n \in \mathfrak{p}$$

for every prime ideal $\mathfrak{p}$. Because $\mathfrak{p}$ is prime, this implies

$$a \in \mathfrak{p}.$$

Thus

$$a \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Conversely, suppose a is not nilpotent. Then the multiplicative set

$$S = \{1, a, a^2, \dots\}$$

does not contain 0. By the standard prime ideal existence lemma, there exists a prime ideal $\mathfrak{p}$ disjoint from S. In particular,

$$a \notin \mathfrak{p}.$$

Therefore a is not contained in the intersection of all prime ideals. Hence

$$\mathrm{Nil}(A) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

□

Corollary 2.3. *For $f \in A$,*

$$D(f) = \varnothing$$

if and only if f is nilpotent.

Proof. We have

$$D(f) = \varnothing$$

if and only if no prime ideal avoids f. Equivalently,

$$f \in \mathfrak{p}$$

for every $\mathfrak{p} \in \operatorname{Spec} A$. Hence

$$f \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

By the previous theorem,

$$f \in \operatorname{Nil}(A).$$

Thus f is nilpotent.

Conversely, if f is nilpotent, then every prime ideal contains f, so no prime ideal lies in $D(f)$. Hence

$$D(f) = \varnothing.$$

□

2.6 Nilpotents Are Topologically Invisible

Let

$$A_{\text{red}} = A/\operatorname{Nil}(A)$$

be the reduced quotient of A.

Proposition 2.2. *There is a natural homeomorphism*

$$\operatorname{Spec} A \cong \operatorname{Spec} A_{\text{red}}.$$

Proof. Prime ideals of $A/\operatorname{Nil}(A)$ correspond to prime ideals of A containing $\operatorname{Nil}(A)$. But every prime ideal of A contains $\operatorname{Nil}(A)$. Therefore the prime ideals of A and A_{red} are naturally the same, and the Zariski topologies agree. □

Thus nilpotent elements are not visible in the underlying topological space. They are visible only in the richer scheme structure.

3 Understanding Points of $\operatorname{Spec} A$

3.1 Prime Ideals as Points

A point of $\operatorname{Spec} A$ is not necessarily a classical point. It is a prime ideal.

For example, in

$$\operatorname{Spec} \mathbb{C}[x, y],$$

the maximal ideals

$$(x - a, y - b), \qquad (a, b) \in \mathbb{C}^2,$$

are the ordinary closed points of the affine plane.

But $\operatorname{Spec} \mathbb{C}[x, y]$ also contains points such as

$$(x)$$

and

$$(y - x^2).$$

These are not ordinary points. They are generic points of irreducible curves.

It also contains

$$(0),$$

which is the generic point of the whole plane.

3.2 Closure of a Point

Proposition 3.1. *If* $\mathfrak{p} \in \operatorname{Spec} A$*, then*

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Proof. The smallest closed set containing $\mathfrak{p}$ is the intersection of all closed sets $V(I)$ such that $\mathfrak{p} \in V(I)$. The condition $\mathfrak{p} \in V(I)$ means

$$I \subseteq \mathfrak{p}.$$

The smallest such closed set is therefore

$$V(\mathfrak{p}).$$

□

Therefore

$$\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$$

if and only if

$$\mathfrak{p} \subseteq \mathfrak{q}.$$

Thus smaller prime ideals are more generic, and larger prime ideals are more special.

3.3 Specialization Chains

A chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

corresponds to a specialization chain

$$\mathfrak{p}_0 \rightsquigarrow \mathfrak{p}_1 \rightsquigarrow \cdots \rightsquigarrow \mathfrak{p}_n.$$

For example, in $\mathbb{C}[x, y]$,

$$(0) \subset (x) \subset (x, y).$$

Geometrically this means:

$$\text{generic point of the plane} \rightsquigarrow \text{generic point of the line } x = 0 \rightsquigarrow \text{closed point } (0, 0).$$

$$
\begin{tikzcd}
\boxed{\begin{array}{c}(0)\\ \text{generic point}\\ \text{of the plane}\end{array}} \arrow[d] \\
\boxed{\begin{array}{c}(x)\\ \text{generic point}\\ \text{of the line } x=0\end{array}} \arrow[d] \\
\boxed{\begin{array}{c}(x,y)\\ \text{closed point}\\ (0,0)\end{array}}
\end{tikzcd}
$$

3.4 Closed Points

Proposition 3.2. *A point* $\mathfrak{p} \in \operatorname{Spec} A$ *is closed if and only if* $\mathfrak{p}$ *is a maximal ideal.*

Proof. The closure of $\{\mathfrak{p}\}$ is

$$V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} A \mid \mathfrak{p} \subseteq \mathfrak{q}\}.$$

This equals $\{\mathfrak{p}\}$ if and only if there is no prime ideal strictly containing $\mathfrak{p}$. This is equivalent to $\mathfrak{p}$ being maximal. □

Thus

$$\boxed{\text{closed points of } \operatorname{Spec} A = \text{maximal ideals of } A.}$$

4 Dimension and Spectra of Polynomial Rings

4.1 Krull Dimension

Definition 4.1. *The Krull dimension of* A*, denoted* $\dim A$*, is the supremum of the lengths of chains of prime ideals*

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

Examples:

$$\dim k[x] = 1,$$

because there are chains

$$(0) \subset (x - a).$$

Also,

$$\dim k[x, y] = 2,$$

because there are chains

$$(0) \subset (x) \subset (x, y).$$

Similarly,

$$\dim \mathbb{Z}[x] = 2,$$

because there are chains

$$(0) \subset (p) \subset (p, x).$$

4.2 Spec $k[x]$

Let k be a field. Since $k[x]$ is a PID, its prime ideals are

$$(0)$$

and

$$(f),$$

where $f \in k[x]$ is irreducible.

The point (0) is generic:

$$\overline{\{(0)\}} = \operatorname{Spec} k[x].$$

Every nonzero prime ideal is maximal, so every (f), with f irreducible, is a closed point.

If $k = \mathbb{C}$, every irreducible polynomial is linear. Hence the closed points are

$$(x - a), \qquad a \in \mathbb{C}.$$

If $k = \mathbb{R}$, irreducible polynomials are linear or quadratic with negative discriminant. Hence closed points are of the form

$$(x - a), \qquad a \in \mathbb{R},$$

and

$$(q(x)),$$

where $q(x)$ is irreducible quadratic over $\mathbb{R}$.

If $k = \mathbb{F}_p$, closed points correspond to irreducible polynomials over $\mathbb{F}_p$. A polynomial of degree d gives a residue field

$$\mathbb{F}_p[x]/(f) \cong \mathbb{F}_{p^d}.$$

4.3 Spec $\mathbb{C}[x, y]$

The ring $\mathbb{C}[x, y]$ has dimension 2. Its prime ideals have height 0, 1, or 2.

The height 0 prime is

$$(0).$$

This is the generic point of the whole plane.

The height 1 prime ideals are

$$(f),$$

where $f \in \mathbb{C}[x, y]$ is irreducible. These are generic points of irreducible curves.

The height 2 primes are maximal ideals. By Hilbert's Nullstellensatz, they are exactly

$$(x - a, y - b), \qquad (a, b) \in \mathbb{C}^2.$$

These are the closed points.

Thus $\operatorname{Spec} \mathbb{C}[x, y]$ contains:

Prime ideal	Geometric meaning	Closure
(0)	generic point of the plane	$\operatorname{Spec} \mathbb{C}[x, y]$
(f)	generic point of curve $f = 0$	$V(f)$
$(x - a, y - b)$	closed point (a, b)	$\{(x - a, y - b)\}$

4.4 $\operatorname{Spec} \mathbb{Z}[x]$

The ring $\mathbb{Z}[x]$ has dimension 2. It behaves like an arithmetic surface.

There is a generic point:

$$(0).$$

There are height 1 primes of two main types.

First, for every prime number p, the ideal

$$(p)$$

is prime, because

$$\mathbb{Z}[x]/(p) \cong \mathbb{F}_p[x]$$

is a domain. These are called vertical curves.

Second, if $F(x) \in \mathbb{Z}[x]$ is primitive and irreducible, then

$$(F(x))$$

is a height 1 prime. These are called horizontal curves.

The closed points are maximal ideals of the form

$$(p, \overline{f}(x)),$$

where p is a prime number and $\overline{f}(x) \in \mathbb{F}_p[x]$ is irreducible.

Equivalently, using a lift $f(x) \in \mathbb{Z}[x]$, we may write such a maximal ideal as

$$(p, f(x)).$$

Its residue field is

$$\mathbb{Z}[x]/(p, f(x)) \cong \mathbb{F}_p[x]/(\overline{f}(x)),$$

a finite field.

A typical chain of primes is

$$(0) \subset (p) \subset (p, x).$$

Another typical chain is

$$(0) \subset (x) \subset (p, x).$$

$$\begin{tikzcd}
 & \begin{array}{c}(0)\\ \text{generic point}\end{array} \arrow[dl] \arrow[dr] & \\
\begin{array}{c}(p)\\ \text{vertical curve}\end{array} \arrow[dr] & & \begin{array}{c}(x)\\ \text{horizontal curve}\end{array} \arrow[dl] \\
 & \begin{array}{c}(p,x)\\ \text{closed point}\end{array} &
\end{tikzcd}$$

5 Connectedness and Idempotents

5.1 Idempotents

Definition 5.1. *An element $e \in A$ is called idempotent if*

$$e^2 = e.$$

The elements 0 and 1 are always idempotent. An idempotent e is called nontrivial if

$$e \neq 0 \quad \textit{and} \quad e \neq 1.$$

Idempotents are algebraic versions of characteristic functions of clopen subsets.

In topology, a clopen subset $U \subseteq X$ has a characteristic function

$$\chi_U : X \to \{0, 1\}.$$

It satisfies

$$\chi_U^2 = \chi_U.$$

Thus it is idempotent.

In affine algebraic geometry, idempotents in A play exactly this role for $\operatorname{Spec} A$.

5.2 Idempotents Give Clopen Sets

Proposition 5.1. *If $e \in A$ is idempotent, then $D(e)$ is both open and closed.*

Proof. The set $D(e)$ is open by definition. We show it is closed.

Since $e^2 = e$, we have

$$e(1-e) = 0.$$

Let $\mathfrak{p} \in \operatorname{Spec} A$. Since $0 \in \mathfrak{p}$, we have

$$e(1-e) \in \mathfrak{p}.$$

Because $\mathfrak{p}$ is prime,

$$e \in \mathfrak{p} \quad \text{or} \quad 1-e \in \mathfrak{p}.$$

Both cannot occur, because then

$$1 = e + (1-e) \in \mathfrak{p},$$

which is impossible.

Therefore, for every prime ideal $\mathfrak{p}$, exactly one of e and $1-e$ belongs to $\mathfrak{p}$. Hence

$$e \notin \mathfrak{p}$$

if and only if

$$1-e \in \mathfrak{p}.$$

Thus

$$D(e) = V(1-e).$$

So $D(e)$ is closed. □

5.3 Clopen Sets Come from Idempotents

Theorem 5.1. *Clopen subsets of* $\operatorname{Spec} A$ *correspond exactly to idempotents of A. More precisely, if $e \in A$ is idempotent, then $D(e)$ is clopen, and every clopen subset of* $\operatorname{Spec} A$ *is of the form $D(e)$ for some idempotent $e \in A$.*

This theorem is one of the clearest bridges between algebra and topology:

$$\boxed{\text{idempotents in } A \quad \longleftrightarrow \quad \text{clopen subsets of } \operatorname{Spec} A.}$$

5.4 Disconnected Spectra and Product Rings

Theorem 5.2. *For a ring A, the following are equivalent:*

i) $\operatorname{Spec} A$ *is disconnected.*

ii) A has a nontrivial idempotent.

iii) There exist nonzero orthogonal idempotents $e_1, e_2 \in A$ such that

$$e_1 e_2 = 0, \qquad e_1 + e_2 = 1.$$

iv) $A \cong A_1 \times A_2$ *for some nonzero rings* A_1, A_2.

Proof. Suppose first that $A \cong A_1 \times A_2$. Then

$$e_1 = (1,0), \qquad e_2 = (0,1)$$

are nonzero idempotents satisfying

$$e_1 e_2 = 0, \qquad e_1 + e_2 = 1.$$

Conversely, suppose $e_1, e_2 \in A$ satisfy

$$e_1 e_2 = 0, \qquad e_1^2 = e_1, \qquad e_2^2 = e_2, \qquad e_1 + e_2 = 1.$$

Define

$$A_1 = Ae_1, \qquad A_2 = Ae_2.$$

Then

$$A \cong Ae_1 \times Ae_2$$

by the map

$$a \longmapsto (ae_1, ae_2).$$

Now consider the topology. If e is a nontrivial idempotent, then

$$D(e)$$

and

$$D(1-e)$$

are disjoint nonempty clopen subsets, and

$$\operatorname{Spec} A = D(e) \sqcup D(1-e).$$

Thus $\operatorname{Spec} A$ is disconnected.

Conversely, if $\operatorname{Spec} A$ is disconnected, then it has a nonempty proper clopen subset U. By the idempotent–clopen correspondence, there exists an idempotent $e \in A$ such that

$$U = D(e).$$

Since U is nonempty and proper, e is nontrivial. Hence A has a nontrivial idempotent. □

Corollary 5.3.

$$\operatorname{Spec} A \textit{ is connected}$$

if and only if the only idempotents of A *are*

$$0 \quad \textit{and} \quad 1.$$

6 Examples

6.1 Example 1: $\operatorname{Spec} k[x]$

Let k be a field and

$$A = k[x].$$

Since $k[x]$ is a PID, the prime ideals are

$$(0)$$

and

$$(f),$$

where $f \in k[x]$ is irreducible.

The closure of the generic point is

$$\overline{\{(0)\}} = \operatorname{Spec} k[x].$$

Every nonzero prime ideal is maximal, so every (f) with f irreducible is a closed point.

If $k = \mathbb{C}$, then the closed points are

$$(x - a), \qquad a \in \mathbb{C}.$$

If $k = \mathbb{R}$, then the closed points are

$$(x - a), \qquad a \in \mathbb{R},$$

and

$$(q(x)),$$

where $q(x)$ is irreducible quadratic over $\mathbb{R}$.

If $k = \mathbb{F}_p$, then closed points correspond to irreducible polynomials over $\mathbb{F}_p$. An irreducible polynomial of degree d gives a residue field

$$\mathbb{F}_p[x]/(f) \cong \mathbb{F}_{p^d}.$$

6.2 Example 2: $D(x) \subseteq \operatorname{Spec} k[x]$

Let

$$A = k[x].$$

Then

$$D(x) = \{\mathfrak{p} \in \operatorname{Spec} k[x] \mid x \notin \mathfrak{p}\}.$$

The closed point (x) is excluded, because

$$x \in (x).$$

The generic point (0) is included, because

$$x \notin (0).$$

The closed point $(x - a)$ is included if and only if

$$a \neq 0.$$

Therefore $D(x)$ is the affine line with the origin removed:

$$D(x) = \mathbb{A}^1_k \setminus \{0\}.$$

Algebraically,

$$D(x) \cong \operatorname{Spec} k[x, x^{-1}].$$

Thus removing the zero set of x corresponds to inverting x.

6.3 Example 3: $D(xy) \subseteq \operatorname{Spec} k[x, y]$

Let

$$A = k[x, y].$$

Then

$$D(xy) = D(x) \cap D(y).$$

Geometrically,

$$D(x)$$

is the plane with the y-axis removed, while

$$D(y)$$

is the plane with the x-axis removed.

Thus

$$D(xy)$$

is the affine plane with both coordinate axes removed.

Algebraically,

$$D(xy) \cong \operatorname{Spec} k[x, y, (xy)^{-1}].$$

Since inverting xy forces both x and y to become invertible, this is also

$$\operatorname{Spec} k[x, x^{-1}, y, y^{-1}].$$

6.4 Example 4: Nilpotent Thickening $k[\varepsilon]/(\varepsilon^2)$

Let

$$A = k[\varepsilon]/(\varepsilon^2).$$

Here

$$\varepsilon^2 = 0,$$

so ε is nilpotent.

The nilradical is

$$\mathrm{Nil}(A) = (\varepsilon).$$

The reduced quotient is

$$A_{\mathrm{red}} = A/(\varepsilon) \cong k.$$

Therefore

$$\operatorname{Spec} A \cong \operatorname{Spec} k$$

as topological spaces.

Thus $\operatorname{Spec} A$ has one point:

$$(\varepsilon).$$

Also,

$$D(\varepsilon) = \varnothing,$$

because ε is nilpotent.

This example shows that nilpotents are invisible to the topology of $\operatorname{Spec} A$, although they are visible in the scheme structure.

6.5 Example 5: Disconnected Spectrum $k \times k$

Let

$$A = k \times k.$$

The prime ideals are

$$(0) \times k$$

and

$$k \times (0).$$

Therefore

$$\operatorname{Spec}(k \times k)$$

has two points.

The idempotents are

$$(0,0), \qquad (1,1), \qquad (1,0), \qquad (0,1).$$

The nontrivial idempotents

$$e_1 = (1,0), \qquad e_2 = (0,1)$$

satisfy

$$e_1e_2 = 0, \qquad e_1 + e_2 = 1.$$

Hence

$$\operatorname{Spec}(k \times k) = D(e_1) \sqcup D(e_2),$$

a disjoint union of two points.

6.6 Example 6: $k[x]/(x(x-1))$

Let

$$A = k[x]/(x(x-1)).$$

The ideals (x) and $(x-1)$ are comaximal in $k[x]$. By the Chinese remainder theorem,

$$k[x]/(x(x-1)) \cong k[x]/(x) \times k[x]/(x-1).$$

Since

$$k[x]/(x) \cong k$$

and

$$k[x]/(x-1) \cong k,$$

we get

$$A \cong k \times k.$$

Therefore

$$\operatorname{Spec} A \cong \operatorname{Spec} k \sqcup \operatorname{Spec} k.$$

So $\operatorname{Spec} A$ is disconnected.

Inside A, the two nontrivial idempotents are represented by

$$e_1 = 1 - x, \qquad e_2 = x.$$

Indeed, in the quotient A, we have

$$x(x-1) = 0.$$

Thus

$$x^2 = x.$$

Therefore

$$e_2^2 = x^2 = x = e_2.$$

Also,

$$e_1^2 = (1-x)^2 = 1 - 2x + x^2.$$

Since $x^2 = x$, this becomes

$$e_1^2 = 1 - 2x + x = 1 - x = e_1.$$

Moreover,

$$e_1e_2 = (1-x)x = x - x^2 = 0,$$

and

$$e_1 + e_2 = (1-x) + x = 1.$$

Thus the disconnectedness of $\operatorname{Spec} A$ is encoded algebraically by the idempotents x and $1-x$.

6.7 Example 7: $\mathbb{Z}/6\mathbb{Z}$

Let

$$A = \mathbb{Z}/6\mathbb{Z}.$$

Since

$$6 = 2 \cdot 3$$

and the ideals (2) and (3) are comaximal, the Chinese remainder theorem gives

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

Therefore

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3.$$

Hence

$$\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z}) \cong \operatorname{Spec} \mathbb{F}_2 \sqcup \operatorname{Spec} \mathbb{F}_3.$$

So $\operatorname{Spec}(\mathbb{Z}/6\mathbb{Z})$ consists of two disconnected points.

The two nontrivial orthogonal idempotents are

$$3 \pmod 6 \qquad \text{and} \qquad 4 \pmod 6.$$

Indeed,

$$3^2 = 9 \equiv 3 \pmod 6,$$

and

$$4^2 = 16 \equiv 4 \pmod 6.$$

Also,

$$3 \cdot 4 = 12 \equiv 0 \pmod 6,$$

and

$$3 + 4 = 7 \equiv 1 \pmod 6.$$

Thus

$$e_1 = 3, \qquad e_2 = 4$$

give the decomposition.

6.8 Example 8: Reducible but Connected, $k[x,y]/(xy)$

Let

$$A = k[x,y]/(xy).$$

Geometrically, this is the union of the two coordinate axes:

$$xy = 0.$$

The two irreducible components are

$$V(x)$$

and

$$V(y).$$

They meet at the origin:

$$(x,y).$$

Therefore the space is reducible but connected.

The minimal prime ideals are

$$(x) \quad \text{and} \quad (y).$$

They correspond to the two irreducible components.

However, the components are not disjoint, because both contain the closed point

$$(x,y).$$

Thus $\operatorname{Spec} k[x,y]/(xy)$ is connected.

Equivalently, the ring

$$k[x,y]/(xy)$$

has no nontrivial idempotents.

This example shows that reducible does not imply disconnected.

$$\boxed{\text{reducible} \not\Rightarrow \text{disconnected.}}$$

6.9 Example 9: Comparing Two Reducible Rings

Compare

$$A_1 = k[x,y]/(xy)$$

and

$$A_2 = k[x]/(x(x-1)).$$

For

$$A_1 = k[x,y]/(xy),$$

the geometric picture is two lines crossing. The components meet at the origin, so the spectrum is connected.

For
$$A_2 = k[x]/(x(x-1)),$$
the geometric picture is two distinct points $x = 0$ and $x = 1$. They do not meet, so the spectrum is disconnected.

Thus:
$$k[x,y]/(xy) \quad \text{is reducible but connected,}$$
whereas
$$k[x]/(x(x-1)) \quad \text{is reducible and disconnected.}$$

The difference is intersection.

6.10 Example 10: $\operatorname{Spec}\mathbb{Z}$

The prime ideals of $\mathbb{Z}$ are
$$(0)$$
and
$$(p),$$
where p is a prime number.

The point (0) is generic:
$$\overline{\{(0)\}} = \operatorname{Spec}\mathbb{Z}.$$

Each (p) is closed.

The space looks like one generic point together with closed points indexed by prime numbers.

Although the closed points are separated from each other, the whole space is connected because of the generic point.

Algebraically, $\mathbb{Z}$ is a domain, and a domain has no nontrivial idempotents. Indeed, if
$$e^2 = e,$$
then
$$e(e-1) = 0.$$
In a domain, this implies
$$e = 0 \quad \text{or} \quad e = 1.$$

Therefore
$$\operatorname{Spec}\mathbb{Z}$$
is connected.

6.11 Example 11: $\operatorname{Spec}\mathbb{Z}[x]$

The spectrum

$$\operatorname{Spec}\mathbb{Z}[x]$$

has dimension 2.

There are chains

$$(0) \subset (p) \subset (p, x),$$

and

$$(0) \subset (x) \subset (p, x).$$

The prime ideal

$$(p)$$

is a vertical curve because

$$\mathbb{Z}[x]/(p) \cong \mathbb{F}_p[x].$$

The prime ideal

$$(x)$$

is a horizontal curve because

$$\mathbb{Z}[x]/(x) \cong \mathbb{Z}.$$

The maximal ideal

$$(p, x)$$

is a closed point, and its residue field is

$$\mathbb{Z}[x]/(p, x) \cong \mathbb{F}_p.$$

Thus $\operatorname{Spec}\mathbb{Z}[x]$ is a two-dimensional arithmetic space mixing geometry and number theory.

6.12 Example 12: Specialization in $\operatorname{Spec}\mathbb{C}[x, y]$

Consider the chain

$$(0) \subset (y - x^2) \subset (x, y).$$

The ideal

$$(0)$$

is the generic point of the whole affine plane.

The ideal

$$(y - x^2)$$

is the generic point of the parabola

$$y = x^2.$$

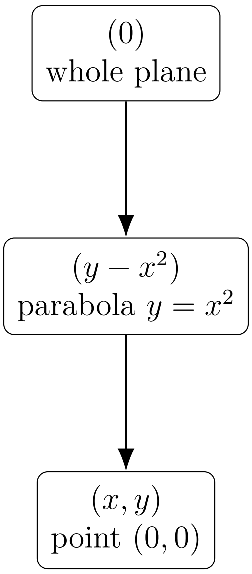
The ideal

$$(x, y)$$

is the closed point $(0, 0)$.

Thus the specialization chain is:

$$\text{plane} \rightsquigarrow \text{parabola} \rightsquigarrow \text{point}.$$



This example shows why $\operatorname{Spec} \mathbb{C}[x, y]$ contains more than ordinary points: it also contains generic points of irreducible subvarieties.

7 Summary Tables

7.1 Basic Dictionary

Algebra	Geometry/topology
A	$\operatorname{Spec} A$
$f \in A$	function on $\operatorname{Spec} A$
$D(f)$	open set where $f \neq 0$
A_f	coordinate ring of $D(f)$
$\mathfrak{p} \in \operatorname{Spec} A$	point
$\mathfrak{m}$ maximal	closed point
$\mathfrak{p}$ nonmaximal	generic point
$\operatorname{Nil}(A)$	nilpotents invisible topologically
$e^2 = e$	clopen decomposition
$A_1 \times A_2$	disjoint union of spectra

7.2 Important Examples

A	$\operatorname{Spec} A$	Connected?	Reason
k	one point	yes	field
$k[x]$	affine line with generic point	yes	domain
$k[\varepsilon]/(\varepsilon^2)$	one point topologically	yes	nilpotent thickening
$k \times k$	two points	no	product ring
$k[x]/(x(x-1))$	two closed points	no	Chinese remainder theorem
$k[x,y]/(xy)$	two lines crossing	yes	components meet
$\mathbb{Z}$	generic point plus primes	yes	domain
$\mathbb{Z}/6\mathbb{Z}$	two points	no	$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{F}_2 \times \mathbb{F}_3$

7.3 Connectedness Tests

$$\boxed{\operatorname{Spec} A \text{ connected} \iff A \text{ has no nontrivial idempotents.}}$$

$$\boxed{\operatorname{Spec} A \text{ disconnected} \iff A \cong A_1 \times A_2 \text{ with } A_1, A_2 \neq 0.}$$

$$\boxed{A \text{ a domain} \implies \operatorname{Spec} A \text{ connected.}}$$

The last implication holds because a domain has no nontrivial idempotents.

7.4 Exercise Set

Exercise 7.1. *Describe* $\operatorname{Spec} k[x]$, *where* k *is a field.*

Exercise 7.2. *Describe* $D(x) \subseteq \operatorname{Spec} k[x]$.

Exercise 7.3. *Show that*

$$D(x) \cong \operatorname{Spec} k[x, x^{-1}].$$

Exercise 7.4. *Compute*

$$D(x) \cap D(y)$$

inside $\operatorname{Spec} k[x,y]$.

Exercise 7.5. *Describe*

$$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2).$$

Exercise 7.6. *Show that*

$$D(\varepsilon) = \varnothing$$

in

$$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2).$$

Exercise 7.7. *Compare topologically*

$$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$$

and

$$\operatorname{Spec} k.$$

Exercise 7.8. *Show that*
$$\operatorname{Spec}(k \times k)$$
is disconnected.

Exercise 7.9. *Find all idempotents in*
$$k \times k.$$

Exercise 7.10. *Show that*
$$k[x]/(x(x-1)) \cong k \times k.$$

Exercise 7.11. *Show that*
$$\operatorname{Spec} k[x,y]/(xy)$$
is connected but reducible.

Exercise 7.12. *Show that*
$$\operatorname{Spec} \mathbb{Z}/6\mathbb{Z}$$
is disconnected.

7.5 Final Conceptual Summary

The basic open sets $D(f)$ are the affine local pieces of $\operatorname{Spec} A$. They are controlled by localization:
$$D(f) \cong \operatorname{Spec} A_f.$$

Nilpotents vanish at every prime ideal:
$$D(f) = \varnothing \iff f \in \operatorname{Nil}(A).$$

Thus nilpotents are invisible to the underlying topology.

The points of $\operatorname{Spec} A$ are prime ideals, not only maximal ideals. Maximal ideals are closed points, while nonmaximal prime ideals are generic points of irreducible closed subsets.

Finally, connectedness of $\operatorname{Spec} A$ is controlled by idempotents:
$$\operatorname{Spec} A \text{ disconnected} \iff A \text{ has a nontrivial idempotent} \iff A \cong A_1 \times A_2.$$

So the three central messages are:
$$\boxed{D(f) \text{ means invert } f.}$$
$$\boxed{\operatorname{Spec} A \text{ remembers irreducible geometry through generic points.}}$$
$$\boxed{\text{Disconnected affine geometry is exactly product decomposition of rings.}}$$

Exercise 4

Let $\mathcal{F}$ be a presheaf of abelian groups on a topological space X. Complete the proof of the existence of the associated sheaf $\mathcal{F}^+$ by showing:

1. $\mathcal{F}^+$ is a sheaf.
2. There is a natural morphism of presheaves
$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+$$
such that the pair $(\mathcal{F}^+, \theta)$ satisfies the desired universal property.

Further, show that
$$(\mathcal{F}^+)_p = \mathcal{F}_p \qquad \text{for every } p \in X.$$

Show that if
$$f : \mathcal{F} \longrightarrow \mathcal{G}$$
is a morphism of presheaves, then there is an induced morphism
$$f^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}^+$$
with
$$(f^+)_p = f_p.$$

The universal property can be rephrased in terms of adjoint functors. Let PreSh_X denote the category of presheaves of abelian groups on X, and let Sh_X denote the category of sheaves of abelian groups on X. There is an obvious inclusion functor
$$i : \mathrm{Sh}_X \longrightarrow \mathrm{PreSh}_X\,.$$
The above construction tells us that i has an adjoint functor given by
$$a(\mathcal{F}) = \mathcal{F}^+.$$
This means that, for $\mathcal{F}$ a presheaf and $\mathcal{G}$ a sheaf, there are natural bijections
$$\mathrm{Hom}_{\mathrm{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PreSh}_X}(\mathcal{F}, i\mathcal{G}).$$

Solution

We construct the sheafification of $\mathcal{F}$ by using stalks.

For each point $p \in X$, let
$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U)$$
be the stalk of $\mathcal{F}$ at p.

Elements of $\mathcal{F}_p$ are germs. If $s \in \mathcal{F}(U)$ and $p \in U$, we denote its germ at p by
$$s_p \in \mathcal{F}_p.$$

Define the disjoint union

$$E_{\mathcal{F}} := \coprod_{p \in X} \mathcal{F}_p.$$

There is a natural projection

$$\pi : E_{\mathcal{F}} \longrightarrow X, \qquad \pi(a) = p \quad \text{if } a \in \mathcal{F}_p.$$

For every open set $U \subseteq X$ and every section $s \in \mathcal{F}(U)$, define a map

$$\widetilde{s} : U \longrightarrow E_{\mathcal{F}}$$

by

$$\widetilde{s}(p) = s_p.$$

We topologize $E_{\mathcal{F}}$ by declaring the sets

$$\widetilde{s}(U) = \{s_p : p \in U\}$$

to be a basis of open sets.

Then define

$$\mathcal{F}^+(U)$$

to be the set of all continuous sections of π over U, i.e.

$$\mathcal{F}^+(U) = \{\sigma : U \to E_{\mathcal{F}} \mid \pi \circ \sigma = \mathrm{id}_U,\ \sigma \text{ is continuous}\}.$$

For an inclusion $V \subseteq U$, the restriction map is ordinary restriction:

$$\rho^U_V : \mathcal{F}^+(U) \longrightarrow \mathcal{F}^+(V), \qquad \sigma \longmapsto \sigma|_V.$$

Thus $\mathcal{F}^+$ is at least a presheaf.

1. The presheaf $\mathcal{F}^+$ is a sheaf

Let $U \subseteq X$ be open and let $\{U_i\}_{i \in I}$ be an open cover of U.

Suppose we have sections

$$\sigma_i \in \mathcal{F}^+(U_i)$$

such that

$$\sigma_i|_{U_i \cap U_j} = \sigma_j|_{U_i \cap U_j}$$

for all i, j.

We define a map

$$\sigma : U \longrightarrow E_{\mathcal{F}}$$

by setting

$$\sigma(p) = \sigma_i(p) \quad \text{if } p \in U_i.$$

This is well-defined because if $p \in U_i \cap U_j$, then

$$\sigma_i(p) = \sigma_j(p).$$

Also
$$\pi(\sigma(p)) = p,$$
because each σ_i is a section of π.

It remains to show that σ is continuous.

Continuity is local on the domain. Since each σ_i is continuous and
$$\sigma|_{U_i} = \sigma_i,$$
it follows that σ is continuous.

Hence
$$\sigma \in \mathcal{F}^+(U).$$

Moreover, σ is the unique section with
$$\sigma|_{U_i} = \sigma_i$$
for every i, because the U_i cover U.

Therefore $\mathcal{F}^+$ satisfies the sheaf gluing axiom.

The locality axiom is also immediate: if
$$\sigma, \tau \in \mathcal{F}^+(U)$$
and
$$\sigma|_{U_i} = \tau|_{U_i}$$
for every i, then for every $p \in U$ there exists i with $p \in U_i$, so
$$\sigma(p) = \tau(p).$$

Thus $\sigma = \tau$.

Therefore $\mathcal{F}^+$ is a sheaf.

2. The natural morphism $\theta : \mathcal{F} \to \mathcal{F}^+$

For every open set $U \subseteq X$, define
$$\theta_U : \mathcal{F}(U) \longrightarrow \mathcal{F}^+(U)$$
by
$$\theta_U(s) = \widetilde{s},$$
where
$$\widetilde{s}(p) = s_p.$$

This is well-defined because $\widetilde{s}$ is continuous by construction of the topology on $E_{\mathcal{F}}$.

The maps θ_U are compatible with restrictions. Indeed, if $V \subseteq U$, then for $s \in \mathcal{F}(U)$ and $p \in V$,
$$\theta_V(s|_V)(p) = (s|_V)_p = s_p = \theta_U(s)(p).$$

Hence

$$\theta_V(s|_V) = \theta_U(s)|_V.$$

Thus θ is a morphism of presheaves:

$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+.$$

3. The stalks of $\mathcal{F}^+$

We now show that

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

Define

$$\Phi_p : (\mathcal{F}^+)_p \longrightarrow \mathcal{F}_p$$

by

$$\Phi_p(\sigma_p) = \sigma(p).$$

This makes sense because if

$$\sigma \in \mathcal{F}^+(U)$$

and $p \in U$, then

$$\sigma(p) \in \mathcal{F}_p.$$

We show that Φ_p is an isomorphism.

Surjectivity

Let

$$a \in \mathcal{F}_p.$$

Then a is represented by some section

$$s \in \mathcal{F}(U)$$

with $p \in U$, so

$$a = s_p.$$

Consider

$$\widetilde{s} \in \mathcal{F}^+(U).$$

Then

$$\Phi_p((\widetilde{s})_p) = \widetilde{s}(p) = s_p = a.$$

Thus Φ_p is surjective.

Injectivity

Suppose

$$\sigma, \tau \in \mathcal{F}^+(U)$$

and suppose their germs at p satisfy

$$\Phi_p(\sigma_p) = \Phi_p(\tau_p).$$

That means

$$\sigma(p) = \tau(p) \quad \text{in } \mathcal{F}_p.$$

Because σ and τ are continuous sections of the espace étalé, there exist neighborhoods $V, W \subseteq U$ of p and presheaf sections

$$s \in \mathcal{F}(V), \qquad t \in \mathcal{F}(W),$$

such that

$$\sigma|_V = \widetilde{s}, \qquad \tau|_W = \widetilde{t}.$$

The equality

$$\sigma(p) = \tau(p)$$

means

$$s_p = t_p \quad \text{in } \mathcal{F}_p.$$

Therefore there exists an open neighborhood

$$N \subseteq V \cap W$$

of p such that

$$s|_N = t|_N.$$

Consequently

$$\sigma|_N = \widetilde{s}|_N = \widetilde{t}|_N = \tau|_N.$$

Thus $\sigma_p = \tau_p$ in $(\mathcal{F}^+)_p$.

Therefore Φ_p is injective.

Hence

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

Under this identification,

$$\theta_p : \mathcal{F}_p \longrightarrow (\mathcal{F}^+)_p$$

is the identity map.

So one usually writes

$$(\mathcal{F}^+)_p = \mathcal{F}_p.$$

4. Universal property

Let $\mathcal{G}$ be a sheaf and let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of presheaves.

We must construct a unique morphism of sheaves

$$\varphi^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}$$

such that

$$\varphi^+ \circ \theta = \varphi.$$

Let $U \subseteq X$ be open and let

$$\sigma \in \mathcal{F}^+(U).$$

For each point $p \in U$, the value $\sigma(p)$ lies in $\mathcal{F}_p$. The morphism φ induces a morphism on stalks

$$\varphi_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p.$$

We would like to define a section of $\mathcal{G}$ whose germ at p is

$$\varphi_p(\sigma(p)).$$

This is possible because $\mathcal{G}$ is a sheaf.

Indeed, since σ is continuous, for every $p \in U$ there is an open neighborhood $U_p \subseteq U$ and a section

$$s_p \in \mathcal{F}(U_p)$$

such that

$$\sigma|_{U_p} = \widetilde{s_p}.$$

Then define locally

$$g_p := \varphi_{U_p}(s_p) \in \mathcal{G}(U_p).$$

On overlaps $U_p \cap U_q$, the local sections g_p and g_q have the same germs at every point. Since $\mathcal{G}$ is a sheaf, equality of germs at every point implies equality locally, and therefore the g_p glue uniquely to a section

$$g \in \mathcal{G}(U).$$

Define

$$\varphi_U^+(\sigma) := g.$$

Equivalently, $\varphi_U^+(\sigma)$ is the unique section of $\mathcal{G}(U)$ satisfying

$$\left(\varphi_U^+(\sigma)\right)_p = \varphi_p(\sigma(p)) \qquad \text{for all } p \in U.$$

This construction is compatible with restrictions, so it gives a morphism of sheaves

$$\varphi^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}.$$

Now let $s \in \mathcal{F}(U)$. Then

$$\theta_U(s) = \widetilde{s}.$$

For every $p \in U$,

$$\left(\varphi_U^+(\widetilde{s})\right)_p = \varphi_p(s_p) = (\varphi_U(s))_p.$$

Since $\mathcal{G}$ is a sheaf, sections with the same germs at all points are equal. Hence

$$\varphi_U^+(\theta_U(s)) = \varphi_U(s).$$

Therefore

$$\varphi^+ \circ \theta = \varphi.$$

Finally, uniqueness follows because the local sections $\widetilde{s}$ generate $\mathcal{F}^+$ locally. More explicitly, every section of $\mathcal{F}^+$ is locally of the form $\widetilde{s}$. Therefore any morphism

$$\psi : \mathcal{F}^+ \to \mathcal{G}$$

satisfying

$$\psi \circ \theta = \varphi$$

must agree with φ^+ locally, hence globally by the sheaf axiom.

Thus $(\mathcal{F}^+, \theta)$ satisfies the universal property of sheafification.

5. Functoriality

Let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of presheaves.

Applying the universal property to the morphism

$$\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{\theta_{\mathcal{G}}} \mathcal{G}^+$$

gives a unique morphism of sheaves

$$f^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}^+$$

such that

$$f^+ \circ \theta_{\mathcal{F}} = \theta_{\mathcal{G}} \circ f.$$

On stalks this gives

$$(f^+)_p = f_p.$$

Indeed, under the identifications

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p, \qquad (\mathcal{G}^+)_p \cong \mathcal{G}_p,$$

the induced map is exactly the map on germs

$$f_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p.$$

Therefore sheafification is functorial:

$$\mathcal{F} \longmapsto \mathcal{F}^+, \qquad f \longmapsto f^+.$$

6. Adjointness

Let

$$i : \mathrm{Sh}_X \longrightarrow \mathrm{PreSh}_X$$

be the inclusion functor.

The universal property says exactly that for every presheaf $\mathcal{F}$ and every sheaf $\mathcal{G}$, composition with

$$\theta : \mathcal{F} \to i(\mathcal{F}^+)$$

induces a bijection

$$\mathrm{Hom}_{\mathrm{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PreSh}_X}(\mathcal{F}, i\mathcal{G}).$$

Hence the sheafification functor

$$a : \mathrm{PreSh}_X \longrightarrow \mathrm{Sh}_X, \qquad a(\mathcal{F}) = \mathcal{F}^+$$

is left adjoint to the inclusion functor:

$$a \dashv i.$$

Thus sheafification is the best sheaf approximation to a presheaf.

$$\boxed{\mathrm{Hom}_{\mathrm{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PreSh}_X}(\mathcal{F}, \mathcal{G})}$$

whenever $\mathcal{G}$ is already a sheaf.

Exercise 5

Show that if
$$f : \mathcal{F} \longrightarrow \mathcal{G}$$
is a morphism between sheaves, then the sheaf image $\operatorname{im} f$ can be naturally identified with a subsheaf of $\mathcal{G}$.

Solution

Let
$$f : \mathcal{F} \longrightarrow \mathcal{G}$$
be a morphism of sheaves of abelian groups on X.

For every open set $U \subseteq X$, we have a group homomorphism
$$f_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U).$$

One might try to define the image presheaf by
$$U \longmapsto \operatorname{im}(f_U).$$

This is indeed a presheaf, but it need not be a sheaf.

The reason is that being in the image of f_U is a global condition. A section of $\mathcal{G}(U)$ might be locally in the image of f, but fail to come from a single global section of $\mathcal{F}(U)$.

Therefore the correct sheaf-theoretic image is the sheafification of the presheaf image.

Define the presheaf image
$$\mathrm{Im}^{\mathrm{pre}}(f)$$
by
$$\mathrm{Im}^{\mathrm{pre}}(f)(U) = \operatorname{im}\left(f_U : \mathcal{F}(U) \to \mathcal{G}(U)\right).$$

Then the sheaf image is defined as
$$\operatorname{im} f := \left(\mathrm{Im}^{\mathrm{pre}}(f)\right)^{+}.$$

We claim that $\operatorname{im} f$ can be naturally identified with a subsheaf of $\mathcal{G}$.

1. Explicit description of the image sheaf

Define a subobject
$$\mathcal{H} \subseteq \mathcal{G}$$
as follows.

For every open set $U \subseteq X$, let
$$\mathcal{H}(U) = \left\{ s \in \mathcal{G}(U) \;\middle|\; \begin{array}{l} \text{for every } p \in U, \text{ there exists an open neighborhood} \\ V \subseteq U \text{ of } p \text{ and } t \in \mathcal{F}(V) \text{ such that} \\ s|_V = f_V(t) \end{array} \right\}.$$

In words:

$$\mathcal{H}(U) = \{\text{sections of } \mathcal{G}(U) \text{ which are locally in the image of } f\}.$$

Clearly

$$\mathcal{H}(U) \subseteq \mathcal{G}(U).$$

We now show that $\mathcal{H}$ is a subsheaf of $\mathcal{G}$.

2. $\mathcal{H}$ is a subpresheaf of $\mathcal{G}$

Let

$$s \in \mathcal{H}(U)$$

and let

$$W \subseteq U$$

be open.

We must show that

$$s|_W \in \mathcal{H}(W).$$

Take any point $p \in W$. Since $s \in \mathcal{H}(U)$, there is an open neighborhood

$$V \subseteq U$$

of p and a section

$$t \in \mathcal{F}(V)$$

such that

$$s|_V = f_V(t).$$

Then

$$V \cap W$$

is an open neighborhood of p in W, and

$$(s|_W)|_{V\cap W} = s|_{V\cap W} = f_{V\cap W}(t|_{V\cap W}).$$

Thus $s|_W$ is locally in the image of f.

Therefore

$$s|_W \in \mathcal{H}(W).$$

So $\mathcal{H}$ is a subpresheaf of $\mathcal{G}$.

3. $\mathcal{H}$ is a sheaf

Let $U \subseteq X$ be open and let $\{U_i\}_{i \in I}$ be an open cover of U.

Suppose that

$$s_i \in \mathcal{H}(U_i)$$

and

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j.

Since $\mathcal{G}$ is a sheaf, there exists a unique section

$$s \in \mathcal{G}(U)$$

such that

$$s|_{U_i} = s_i$$

for every i.

We must show that

$$s \in \mathcal{H}(U).$$

Let $p \in U$. Choose i such that

$$p \in U_i.$$

Since

$$s_i \in \mathcal{H}(U_i),$$

there is an open neighborhood

$$V \subseteq U_i$$

of p and a section

$$t \in \mathcal{F}(V)$$

such that

$$s_i|_V = f_V(t).$$

But

$$s|_V = s_i|_V.$$

Therefore

$$s|_V = f_V(t).$$

Thus s is locally in the image of f at every point $p \in U$.

Hence

$$s \in \mathcal{H}(U).$$

So $\mathcal{H}$ is a sheaf.

Since each $\mathcal{H}(U)$ is a subgroup of $\mathcal{G}(U)$ and restrictions are inherited from $\mathcal{G}$, $\mathcal{H}$ is a subsheaf of $\mathcal{G}$.

4. Identification with the sheaf image

We now show that

$$\mathcal{H} \cong \operatorname{im} f.$$

The presheaf image

$$\operatorname{Im}^{\mathrm{pre}}(f)$$

is a subpresheaf of $\mathcal{H}$, because if

$$s = f_U(t)$$

globally on U, then certainly s is locally in the image of f.

Thus we have a morphism of presheaves

$$\operatorname{Im}^{\mathrm{pre}}(f) \longrightarrow \mathcal{H}.$$

Since $\mathcal{H}$ is a sheaf, the universal property of sheafification gives a unique morphism of sheaves

$$(\operatorname{Im}^{\mathrm{pre}}(f))^{+} \longrightarrow \mathcal{H}.$$

That is,

$$\operatorname{im} f \longrightarrow \mathcal{H}.$$

Conversely, every section of $\mathcal{H}$ is locally in the presheaf image. Therefore it is exactly a section obtained by gluing local sections of the presheaf image. This is precisely what sheafification does.

Hence

$$\operatorname{im} f \cong \mathcal{H}.$$

Therefore

$$\boxed{\operatorname{im} f(U) = \{s \in \mathcal{G}(U) \mid s \text{ is locally of the form } f(t)\}}$$

and this realizes $\operatorname{im} f$ as a subsheaf of $\mathcal{G}$.

5. Stalkwise interpretation

For every point $p \in X$,

$$(\operatorname{im} f)_p = \operatorname{im}(f_p),$$

where

$$f_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p.$$

Indeed, taking stalks commutes with sheafification and with images in this situation.

Thus a germ

$$s_p \in \mathcal{G}_p$$

lies in $(\operatorname{im} f)_p$ precisely when there exists a germ

$$t_p \in \mathcal{F}_p$$

such that

$$s_p = f_p(t_p).$$

So the sheaf image is the subsheaf of $\mathcal{G}$ whose sections are locally images of sections of $\mathcal{F}$.

$$\boxed{\operatorname{im} f \subseteq \mathcal{G}}$$

Exercise 6

Show that a sequence of sheaves

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

is exact if and only if for every $p \in X$, the corresponding sequence of stalks

$$\cdots \longrightarrow (\mathcal{F}_{i-1})_p \xrightarrow{(d_{i-1})_p} (\mathcal{F}_i)_p \xrightarrow{(d_i)_p} (\mathcal{F}_{i+1})_p \longrightarrow \cdots$$

is exact.

Solution

We prove that exactness of sheaves is a local property.

Recall that the sequence is exact at $\mathcal{F}_i$ if

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

as subsheaves of $\mathcal{F}_i$.

We must show that this happens if and only if

$$\operatorname{im}\Big((d_{i-1})_p\Big) = \ker\Big((d_i)_p\Big)$$

for every point $p \in X$.

1. Exactness of sheaves implies exactness of stalks

Assume that the sequence of sheaves is exact.

Thus, for every i,

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

as sheaves.

Taking stalks at p, we get

$$\Big(\operatorname{im}(d_{i-1})\Big)_p = \Big(\ker(d_i)\Big)_p.$$

Now stalks commute with kernels:

$$\Big(\ker(d_i)\Big)_p = \ker\Big((d_i)_p\Big).$$

Also, by the description of sheaf images from Exercise 5,

$$\Big(\operatorname{im}(d_{i-1})\Big)_p = \operatorname{im}\Big((d_{i-1})_p\Big).$$

Therefore

$$\operatorname{im}\Big((d_{i-1})_p\Big) = \ker\Big((d_i)_p\Big).$$

Hence the stalk sequence is exact at $(\mathcal{F}_i)_p$.

Since i and p were arbitrary, the sequence of stalks is exact at every point.

2. Exactness of stalks implies exactness of sheaves

Now assume that for every $p \in X$, the sequence of stalks is exact.

Thus, for every p,

$$\operatorname{im}\Big((d_{i-1})_p\Big) = \ker\Big((d_i)_p\Big).$$

We must prove

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

as sheaves.

First observe that

$$d_i \circ d_{i-1} = 0$$

as a morphism of sheaves.

Indeed, on stalks,

$$(d_i)_p \circ (d_{i-1})_p = 0$$

for every p by exactness of the stalk sequence. Since a morphism of sheaves is zero if and only if it is zero on every stalk, it follows that

$$d_i \circ d_{i-1} = 0.$$

Therefore

$$\operatorname{im}(d_{i-1}) \subseteq \ker(d_i).$$

We need to prove the reverse inclusion.

Let

$$s \in \ker(d_i)(U)$$

for some open set $U \subseteq X$.

This means

$$s \in \mathcal{F}_i(U)$$

and

$$d_i(s) = 0 \quad \text{in } \mathcal{F}_{i+1}(U).$$

We must show that s is locally in the image of d_{i-1}.

Let $p \in U$.

The germ

$$s_p \in (\mathcal{F}_i)_p$$

satisfies

$$(d_i)_p(s_p) = 0.$$

Thus

$$s_p \in \ker\Big((d_i)_p\Big).$$

By exactness of the stalk sequence,

$$s_p \in \operatorname{im}\Big((d_{i-1})_p\Big).$$

Therefore there exists a germ

$$t_p \in (\mathcal{F}_{i-1})_p$$

such that

$$(d_{i-1})_p(t_p) = s_p.$$

Choose an open neighborhood

$$V \subseteq U$$

of p and a section

$$t \in \mathcal{F}_{i-1}(V)$$

representing the germ t_p.

Then

$$(d_{i-1}(t))_p = s_p.$$

Equality of germs means that after possibly shrinking V, we have

$$d_{i-1}(t) = s|_V.$$

Thus, for every point $p \in U$, there exists a neighborhood V of p such that

$$s|_V \in \operatorname{im}(d_{i-1,V}).$$

Therefore s is locally in the image of d_{i-1}.

By the description of the sheaf image,

$$s \in \operatorname{im}(d_{i-1})(U).$$

Hence

$$\ker(d_i)(U) \subseteq \operatorname{im}(d_{i-1})(U)$$

for every open set U.

Thus

$$\ker(d_i) \subseteq \operatorname{im}(d_{i-1}).$$

Together with the opposite inclusion, we obtain

$$\ker(d_i) = \operatorname{im}(d_{i-1}).$$

Therefore the sequence of sheaves is exact.

Conclusion

We have shown both implications:

$$\boxed{\text{A sequence of sheaves is exact} \iff \text{the induced sequence of stalks is exact at every point.}}$$

Equivalently, exactness of sheaves is a pointwise condition.

The reason is that sheaves are local objects: a section belongs to a kernel or to an image sheaf precisely when the corresponding statement holds locally, and local statements are detected by germs.

8 Presheaves

The three main ideas are:

1. every presheaf has an associated sheaf;
2. the image of a morphism of sheaves is naturally a subsheaf of the target;
3. exactness of sheaves can be checked on stalks.

All three are governed by the same fundamental principle:

$$\boxed{\text{Sheaves are local objects, and stalks detect local behavior.}}$$

A sheaf is not merely a rule assigning an algebraic object to every open set. It is a rule whose sections are determined by their local behavior and whose compatible local sections glue uniquely.

The purpose of sheafification is to take a presheaf and force it to satisfy this local gluing principle.

The purpose of the sheaf image is to replace the global notion of image by the local notion of image.

The reason exactness can be checked on stalks is that exactness is a local condition for sheaves.

8.1 Presheaves

Let X be a topological space.

Definition 8.1 (Presheaf of abelian groups). *A presheaf of abelian groups on X consists of the following data:*

1. *for every open set $U \subseteq X$, an abelian group $\mathcal{F}(U)$;*
2. *for every inclusion of open sets $V \subseteq U$, a restriction map*
$$\rho^U_V : \mathcal{F}(U) \longrightarrow \mathcal{F}(V),$$
usually written as
$$s \longmapsto s|_V;$$
3. *the restriction maps satisfy*
$$\rho^U_U = \mathrm{id}_{\mathcal{F}(U)}$$
and, whenever $W \subseteq V \subseteq U$,
$$\rho^V_W \circ \rho^U_V = \rho^U_W.$$

Thus a presheaf is a contravariant functor from the category of open subsets of X to the category of abelian groups:

$$\mathcal{F} : \mathrm{Open}(X)^{op} \longrightarrow \mathbf{Ab}.$$

The word "contravariant" means that inclusions go in one direction,

$$V \subseteq U,$$

but restriction maps go in the opposite direction,

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V).$$

A sheaf is a presheaf satisfying two additional axioms: locality and gluing.

Definition 8.2 (Sheaf)**.** *A presheaf $\mathcal{F}$ on X is called a sheaf if for every open set $U \subseteq X$ and every open cover*

$$U = \bigcup_{i \in I} U_i,$$

the following two conditions hold.

Locality. *If $s, t \in \mathcal{F}(U)$ and*

$$s|_{U_i} = t|_{U_i}$$

for every i, then

$$s = t.$$

Gluing. *If we are given sections*

$$s_i \in \mathcal{F}(U_i)$$

such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j, then there exists a unique section

$$s \in \mathcal{F}(U)$$

such that

$$s|_{U_i} = s_i$$

for every i.

The slogan is:

$$\boxed{\text{A sheaf is a presheaf whose compatible local data glue uniquely.}}$$

8.2 Stalks and germs

The stalk is the algebraic object that captures the behavior of a presheaf arbitrarily close to one point.

Definition 8.3 (Stalk)**.** *Let $\mathcal{F}$ be a presheaf on X and let $p \in X$. The stalk of $\mathcal{F}$ at p is*

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U),$$

where the direct limit is taken over all open neighborhoods U of p.

If $s \in \mathcal{F}(U)$ and $p \in U$, the image of s in the stalk $\mathcal{F}_p$ is called the germ of s at p and is denoted by

$$s_p \in \mathcal{F}_p.$$

Two sections $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ have the same germ at p if and only if there exists an open neighborhood

$$W \subseteq U \cap V$$

of p such that

$$s|_W = t|_W.$$

Thus

$$s_p = t_p$$

means that s and t become equal after restricting to a sufficiently small neighborhood of p.

Remark 8.1. *The stalk $\mathcal{F}_p$ forgets global behavior and remembers only local behavior near p.*

8.3 Why stalks are important

For sheaves, stalks detect equality.

Proposition 8.1 (Equality can be checked on stalks)**.** *Let $\mathcal{F}$ be a sheaf on X, let $U \subseteq X$ be open, and let $s, t \in \mathcal{F}(U)$.*

Then

$$s = t$$

if and only if

$$s_p = t_p$$

for every $p \in U$.

Proof. If $s = t$, then clearly $s_p = t_p$ for every $p \in U$.

Conversely, suppose that $s_p = t_p$ for every $p \in U$. For each $p \in U$, equality of germs means that there exists an open neighborhood $V_p \subseteq U$ of p such that

$$s|_{V_p} = t|_{V_p}.$$

The sets V_p cover U. By the locality axiom for the sheaf $\mathcal{F}$, we get

$$s = t.$$

$\square$

Corollary 8.2. *Let*

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves. Then $f = 0$ if and only if

$$f_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p$$

is the zero map for every $p \in X$.

Corollary 8.3. *A morphism of sheaves*

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

is an isomorphism if and only if every stalk map

$$f_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p$$

is an isomorphism.

8.4 Sheafification

A presheaf does not necessarily satisfy locality and gluing. Sheafification is the process of replacing a presheaf by the closest sheaf associated to it.

Definition 8.4 (Sheafification)**.** *Let $\mathcal{F}$ be a presheaf on X. A sheafification of $\mathcal{F}$ is a sheaf $\mathcal{F}^+$ together with a morphism of presheaves*

$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+$$

such that for every sheaf $\mathcal{G}$ and every morphism of presheaves

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G},$$

there exists a unique morphism of sheaves

$$\varphi^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}$$

such that

$$\varphi = \varphi^+ \circ \theta.$$

Equivalently, every map from $\mathcal{F}$ into a sheaf factors uniquely through $\mathcal{F}^+$.

This is expressed by the diagram

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d, "\theta"'] & \mathcal{G} \\ \mathcal{F}^+ \arrow[ur, dashed, "\exists!\,\varphi^+"'] & \end{tikzcd}$$

where $\mathcal{G}$ is a sheaf.

8.5 Construction of sheafification by stalks

Let $\mathcal{F}$ be a presheaf on X.

For every $p \in X$, consider the stalk $\mathcal{F}_p$. Form the disjoint union

$$E_{\mathcal{F}} = \coprod_{p \in X} \mathcal{F}_p.$$

There is a natural projection

$$\pi : E_{\mathcal{F}} \longrightarrow X$$

defined by

$$\pi(a) = p \qquad \text{if } a \in \mathcal{F}_p.$$

If $U \subseteq X$ is open and $s \in \mathcal{F}(U)$, define

$$\widetilde{s} : U \longrightarrow E_{\mathcal{F}}$$

by

$$\widetilde{s}(p) = s_p.$$

The set $\widetilde{s}(U)$ is the family of germs of s over U:

$$\widetilde{s}(U) = \{s_p : p \in U\}.$$

We topologize $E_{\mathcal{F}}$ by declaring all sets of the form $\widetilde{s}(U)$ to be a basis of open sets.

Then define

$$\mathcal{F}^+(U) = \{\sigma : U \to E_{\mathcal{F}} \mid \pi \circ \sigma = \mathrm{id}_U,\ \sigma \text{ is continuous}\}.$$

Thus $\mathcal{F}^+(U)$ is the group of continuous sections of π over U.

This construction is called the espace étalé construction.

8.6 Intuition behind the construction

A section

$$\sigma \in \mathcal{F}^+(U)$$

chooses for every point $p \in U$ a germ

$$\sigma(p) \in \mathcal{F}_p.$$

However, not every arbitrary choice of germs is allowed. The section σ must be continuous with respect to the topology on $E_{\mathcal{F}}$.

This continuity means precisely that locally, σ comes from an actual section of the original presheaf $\mathcal{F}$.

Therefore:

$$\boxed{\mathcal{F}^+(U) = \{\text{families of germs over } U \text{ that are locally represented by sections of } \mathcal{F}\}.}$$

So sheafification does not create arbitrary pointwise data. It creates exactly the locally representable data.

8.7 The canonical morphism

There is a natural morphism of presheaves

$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+.$$

For every open set $U \subseteq X$, define

$$\theta_U : \mathcal{F}(U) \longrightarrow \mathcal{F}^+(U)$$

by

$$\theta_U(s) = \widetilde{s}.$$

That is,

$$\theta_U(s)(p) = s_p.$$

So θ sends a section to its family of germs.

Proposition 8.2. *The maps θ_U define a morphism of presheaves*

$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+.$$

Proof. Let $V \subseteq U$ be open. For $s \in \mathcal{F}(U)$ and $p \in V$, we have

$$\theta_V(s|_V)(p) = (s|_V)_p = s_p = \theta_U(s)(p).$$

Hence

$$\theta_V(s|_V) = \theta_U(s)|_V.$$

Thus the maps θ_U are compatible with restrictions. □

8.8 Sheafification preserves stalks

One of the most important facts about sheafification is that it does not change stalks.

Theorem 8.4. *For every $p \in X$, there is a natural isomorphism*

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

Under this identification, the induced map

$$\theta_p : \mathcal{F}_p \longrightarrow (\mathcal{F}^+)_p$$

is the identity map.

Proof. Define

$$\Phi_p : (\mathcal{F}^+)_p \longrightarrow \mathcal{F}_p$$

by

$$\Phi_p(\sigma_p) = \sigma(p).$$

This is well-defined because if two sections of $\mathcal{F}^+$ have the same germ at p, then they agree on some neighborhood of p, hence have the same value at p.

We show that Φ_p is an isomorphism.

First, let $a \in \mathcal{F}_p$. Choose a representative $s \in \mathcal{F}(U)$ with $p \in U$ such that

$$a = s_p.$$

Then $\widetilde{s} \in \mathcal{F}^+(U)$ and

$$\Phi_p((\widetilde{s})_p) = \widetilde{s}(p) = s_p = a.$$

Thus Φ_p is surjective.

Now suppose

$$\Phi_p(\sigma_p) = \Phi_p(\tau_p).$$

Then

$$\sigma(p) = \tau(p)$$

inside $\mathcal{F}_p$.

Because σ and τ are continuous sections of the espace étalé, there exist neighborhoods V, W of p and sections

$$s \in \mathcal{F}(V), \qquad t \in \mathcal{F}(W),$$

such that

$$\sigma|_V = \widetilde{s}, \qquad \tau|_W = \widetilde{t}.$$

The equality $\sigma(p) = \tau(p)$ means

$$s_p = t_p.$$

Therefore there exists a smaller neighborhood

$$N \subseteq V \cap W$$

of p such that

$$s|_N = t|_N.$$

Thus

$$\sigma|_N = \tau|_N.$$

Hence

$$\sigma_p = \tau_p.$$

So Φ_p is injective.

Therefore

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

□

Remark 8.5. *Sheafification does not change local germs. It only repairs the global gluing behavior of the presheaf.*

8.9 Universal property of sheafification

Theorem 8.6 (Universal property)**.** *Let $\mathcal{F}$ be a presheaf and let $\mathcal{G}$ be a sheaf. For every morphism of presheaves*

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G},$$

there exists a unique morphism of sheaves

$$\varphi^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}$$

such that

$$\varphi = \varphi^+ \circ \theta.$$

Proof. Let $U \subseteq X$ be open and let

$$\sigma \in \mathcal{F}^+(U).$$

For each $p \in U$, the value $\sigma(p)$ is an element of $\mathcal{F}_p$. The morphism φ induces a map on stalks

$$\varphi_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p.$$

We want to define a section $g \in \mathcal{G}(U)$ by requiring

$$g_p = \varphi_p(\sigma(p))$$

for all $p \in U$.

This can be done locally. Since σ is a continuous section of the espace étalé, for every $p \in U$ there exists a neighborhood $U_p \subseteq U$ and a section

$$s_p \in \mathcal{F}(U_p)$$

such that

$$\sigma|_{U_p} = \widetilde{s_p}.$$

Define

$$g_p := \varphi_{U_p}(s_p) \in \mathcal{G}(U_p).$$

On overlaps, these locally defined sections agree. Indeed, their germs agree at every point of the overlap, because both are obtained by applying φ to the same local family of germs. Since $\mathcal{G}$ is a sheaf, equality of germs implies local equality, and the sheaf gluing axiom gives a unique section

$$g \in \mathcal{G}(U)$$

such that

$$g|_{U_p} = g_p$$

locally.

Define

$$\varphi_U^+(\sigma) = g.$$

These maps are compatible with restrictions, hence define a morphism of sheaves

$$\varphi^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}.$$

Now let $s \in \mathcal{F}(U)$. Then

$$\theta_U(s) = \widetilde{s}.$$

For every $p \in U$,

$$(\varphi_U^+(\widetilde{s}))_p = \varphi_p(s_p) = (\varphi_U(s))_p.$$

Since $\mathcal{G}$ is a sheaf, sections with the same germs everywhere are equal. Hence

$$\varphi_U^+(\theta_U(s)) = \varphi_U(s).$$

Therefore

$$\varphi^+ \circ \theta = \varphi.$$

Uniqueness follows because every section of $\mathcal{F}^+$ is locally of the form $\widetilde{s}$. Thus any morphism $\psi : \mathcal{F}^+ \to \mathcal{G}$ satisfying $\psi \circ \theta = \varphi$ must agree with φ^+ locally, hence globally. □

8.10 Sheafification as an adjoint functor

Let

$$\mathrm{PreSh}_X$$

denote the category of presheaves of abelian groups on X, and let

$$\mathrm{Sh}_X$$

denote the category of sheaves of abelian groups on X.

There is an inclusion functor

$$i : \mathrm{Sh}_X \longrightarrow \mathrm{PreSh}_X .$$

This functor simply regards a sheaf as a presheaf.

Sheafification defines a functor

$$a : \mathrm{PreSh}_X \longrightarrow \mathrm{Sh}_X, \qquad a(\mathcal{F}) = \mathcal{F}^+.$$

Theorem 8.7. *The sheafification functor is left adjoint to the inclusion functor:*

$$a \dashv i.$$

Equivalently, for every presheaf $\mathcal{F}$ and every sheaf $\mathcal{G}$, there is a natural bijection

$$\mathrm{Hom}_{\mathrm{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PreSh}_X}(\mathcal{F}, i\mathcal{G}).$$

Proof. This is exactly the universal property of sheafification. A morphism

$$\mathcal{F} \longrightarrow i\mathcal{G}$$

from the presheaf $\mathcal{F}$ to the underlying presheaf of the sheaf $\mathcal{G}$ is equivalent to a unique morphism of sheaves

$$\mathcal{F}^+ \longrightarrow \mathcal{G}.$$

This correspondence is natural in both $\mathcal{F}$ and $\mathcal{G}$. □

Thus:

$$\boxed{\mathcal{F}^+ \text{ is the best sheaf approximation of } \mathcal{F}.}$$

9 Examples:

9.1 Example: constant presheaf and constant sheaf

Let A be an abelian group.

Define the constant presheaf by

$$\mathcal{F}(U) = A$$

for every nonempty open set U, with all restriction maps equal to the identity.

This presheaf is usually not a sheaf.

Suppose U has two disjoint nonempty open components U_1 and U_2. Choose elements $a_1, a_2 \in A$ with $a_1 \neq a_2$. Then we can define local sections

$$s_1 = a_1 \in \mathcal{F}(U_1), \qquad s_2 = a_2 \in \mathcal{F}(U_2).$$

Since $U_1 \cap U_2 = \varnothing$, these sections are automatically compatible on overlaps.

If $\mathcal{F}$ were a sheaf, they would glue to a section of $\mathcal{F}(U)$. But a section of $\mathcal{F}(U)$ is just a single element of A, so it cannot restrict to both a_1 and a_2.

Therefore the constant presheaf is not generally a sheaf.

Its sheafification is the constant sheaf

$$\underline{A}.$$

For each open set U,

$$\underline{A}(U) = \{\text{locally constant functions } U \to A\}.$$

Thus:

$$\boxed{\text{The sheafification of the constant presheaf } A \text{ is the sheaf of locally constant } A\text{-valued functions.}}$$

9.2 Kernels of morphisms of sheaves

Let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves.

The kernel is defined objectwise by

$$(\ker f)(U) = \ker\left(f_U : \mathcal{F}(U) \to \mathcal{G}(U)\right).$$

Proposition 9.1. *The presheaf* $\ker f$ *is already a sheaf.*

Proof. Let $U = \bigcup_i U_i$ be an open cover.

Suppose $s, t \in (\ker f)(U)$ and

$$s|_{U_i} = t|_{U_i}$$

for every i. Since $\mathcal{F}$ is a sheaf, $s = t$. Thus locality holds.

Now suppose $s_i \in (\ker f)(U_i)$ are compatible on overlaps. Since $\mathcal{F}$ is a sheaf, there exists a unique section

$$s \in \mathcal{F}(U)$$

such that

$$s|_{U_i} = s_i.$$

We need to show that $s \in (\ker f)(U)$, i.e.

$$f_U(s) = 0.$$

But

$$f_U(s)|_{U_i} = f_{U_i}(s_i) = 0$$

for every i. Since $\mathcal{G}$ is a sheaf, a section that is locally zero is zero. Therefore

$$f_U(s) = 0.$$

Hence $s \in (\ker f)(U)$. □

So kernels of sheaves are computed objectwise.

$$\boxed{\text{Kernels of sheaf morphisms are computed section by section.}}$$

9.3 Images of morphisms of sheaves

Images behave differently from kernels.

Let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves.

For each open set U, there is a group homomorphism

$$f_U : \mathcal{F}(U) \to \mathcal{G}(U).$$

One might try to define the image presheaf by

$$U \longmapsto \operatorname{im}(f_U).$$

This is indeed a presheaf, denoted

$$\operatorname{Im}^{\text{pre}}(f),$$

where

$$\operatorname{Im}^{\text{pre}}(f)(U) = \operatorname{im}\left(f_U : \mathcal{F}(U) \to \mathcal{G}(U)\right).$$

But this presheaf need not be a sheaf.

The reason is that being globally in the image is stronger than being locally in the image.

A section

$$s \in \mathcal{G}(U)$$

may locally be of the form

$$s|_{U_i} = f(t_i),$$

but there may not exist a single global

$$t \in \mathcal{F}(U)$$

such that

$$s = f(t).$$

Therefore the sheaf image is defined as

$$\operatorname{im} f = \left(\operatorname{Im}^{\mathrm{pre}}(f)\right)^{+}.$$

9.4 The sheaf image as a subsheaf

Although the sheaf image is defined by sheafifying the presheaf image, it has a very concrete description.

Theorem 9.1. *Let*

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves. Then $\operatorname{im} f$ *is naturally identified with the subsheaf* $\mathcal{H}$ *of* $\mathcal{G}$ *defined by*

$$\mathcal{H}(U) = \{ s \in \mathcal{G}(U) \mid s \textit{ is locally in the image of } f \}.$$

Equivalently,

$$\mathcal{H}(U) = \{ s \in \mathcal{G}(U) \mid \forall p \in U,\ \exists V \subseteq U \textit{ open with } p \in V,\ \exists t \in \mathcal{F}(V) \textit{ such that } s|_V = f_V(t) \}.$$

Proof. First observe that $\mathcal{H}(U)$ is a subgroup of $\mathcal{G}(U)$. It is closed under addition and inverses because f_V is a group homomorphism.

If $W \subseteq U$ and $s \in \mathcal{H}(U)$, then $s|_W$ is locally in the image of f. Indeed, if $p \in W$, choose $V \subseteq U$ and $t \in \mathcal{F}(V)$ such that

$$s|_V = f_V(t).$$

Then on $V \cap W$,

$$s|_{V\cap W} = f_{V\cap W}(t|_{V\cap W}).$$

Hence $s|_W \in \mathcal{H}(W)$. So $\mathcal{H}$ is a subpresheaf of $\mathcal{G}$.

Now we prove that $\mathcal{H}$ is a sheaf. Let $U = \bigcup_i U_i$ and suppose

$$s_i \in \mathcal{H}(U_i)$$

are compatible on overlaps. Since $\mathcal{G}$ is a sheaf, there exists a unique

$$s \in \mathcal{G}(U)$$

such that

$$s|_{U_i} = s_i.$$

We must show that $s \in \mathcal{H}(U)$.

Let $p \in U$. Choose i such that $p \in U_i$. Since $s_i \in \mathcal{H}(U_i)$, there exists an open neighborhood

$$V \subseteq U_i$$

of p and a section

$$t \in \mathcal{F}(V)$$

such that

$$s_i|_V = f_V(t).$$

But $s|_V = s_i|_V$, so

$$s|_V = f_V(t).$$

Thus s is locally in the image of f. Hence $s \in \mathcal{H}(U)$. Therefore $\mathcal{H}$ is a subsheaf of $\mathcal{G}$.

The presheaf image

$$\mathrm{Im}^{\mathrm{pre}}(f)$$

maps naturally into $\mathcal{H}$, since a section globally in the image is certainly locally in the image.

By the universal property of sheafification, this map induces a morphism

$$(\mathrm{Im}^{\mathrm{pre}}(f))^+ \longrightarrow \mathcal{H}.$$

Conversely, every section of $\mathcal{H}$ is locally a section of the presheaf image. This is precisely what sheafification adds: compatible locally defined sections. Therefore

$$(\mathrm{Im}^{\mathrm{pre}}(f))^+ \cong \mathcal{H}.$$

Thus

$$\operatorname{im} f \cong \mathcal{H} \subseteq \mathcal{G}.$$

□

Hence:

$$\boxed{(\operatorname{im} f)(U) = \{s \in \mathcal{G}(U) \mid s \text{ is locally of the form } f(t)\}.}$$

9.5 Stalks of image sheaves

Proposition 9.2. *Let*

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves. Then for every $p \in X$,

$$(\operatorname{im} f)_p = \operatorname{im}(f_p),$$

where

$$f_p : \mathcal{F}_p \to \mathcal{G}_p$$

is the induced morphism on stalks.

Proof. A germ in $(\operatorname{im} f)_p$ is represented by a section

$$s \in (\operatorname{im} f)(U)$$

for some neighborhood U of p. By the local description of $\operatorname{im} f$, after shrinking U if necessary, there is a section

$$t \in \mathcal{F}(U)$$

such that

$$s = f_U(t).$$

Therefore

$$s_p = f_p(t_p),$$

so

$$s_p \in \operatorname{im}(f_p).$$

Conversely, if $a \in \operatorname{im}(f_p)$, then there exists

$$b \in \mathcal{F}_p$$

such that

$$a = f_p(b).$$

Choose a representative

$$t \in \mathcal{F}(U)$$

of b. Then

$$f_U(t) \in (\operatorname{im} f)(U),$$

and the germ of $f_U(t)$ at p is a. Thus

$$a \in (\operatorname{im} f)_p.$$

Hence

$$(\operatorname{im} f)_p = \operatorname{im}(f_p).$$

□

9.6 Example: the exponential morphism

Let X be a complex manifold. Let $\mathcal{O}_X$ be the sheaf of holomorphic functions and let $\mathcal{O}_X^\times$ be the sheaf of nowhere-zero holomorphic functions.

There is a morphism of sheaves

$$\exp : \mathcal{O}_X \longrightarrow \mathcal{O}_X^\times$$

given by

$$f \longmapsto e^{2\pi i f}.$$

Locally, every nowhere-zero holomorphic function has a holomorphic logarithm. Therefore locally every section of $\mathcal{O}_X^\times$ is in the image of exp.

Thus

$$\mathrm{im}(\exp) = \mathcal{O}_X^\times$$

as sheaves.

However, globally, a nowhere-zero holomorphic function need not have a global logarithm.

For example, on

$$X = \mathbb{C}^\times,$$

the function

$$z \longmapsto z$$

has no global holomorphic logarithm.

Therefore the map on global sections

$$\mathcal{O}_X(X) \longrightarrow \mathcal{O}_X^\times(X)$$

need not be surjective, even though the morphism of sheaves is surjective.

This illustrates the important distinction:

$$\boxed{\text{Surjective as a sheaf} \neq \text{surjective on global sections.}}$$

9.7 Exactness of sheaves

Let

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

be a sequence of sheaves.

Definition 9.1 (Exactness of sheaves)**.** *The sequence is exact at $\mathcal{F}_i$ if*

$$\mathrm{im}(d_{i-1}) = \ker(d_i)$$

as subsheaves of $\mathcal{F}_i$.

Using the local description of the sheaf image, exactness at $\mathcal{F}_i$ means the following.

If

$$s \in \mathcal{F}_i(U)$$

satisfies

$$d_i(s) = 0,$$

then for every point $p \in U$, there exists an open neighborhood

$$V \subseteq U$$

of p and a section

$$t \in \mathcal{F}_{i-1}(V)$$

such that

$$s|_V = d_{i-1}(t).$$

Therefore:

$$\boxed{\text{Exactness of sheaves means local exactness of sections.}}$$

It does not necessarily mean that every global kernel section is globally an image.

10 Exactness can be checked on stalks

This is one of the most fundamental theorems in elementary sheaf theory.

Theorem 10.1 (Stalkwise criterion for exactness)**.** *A sequence of sheaves*

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

is exact if and only if for every point $p \in X$, the induced sequence of stalks

$$\cdots \longrightarrow (\mathcal{F}_{i-1})_p \xrightarrow{(d_{i-1})_p} (\mathcal{F}_i)_p \xrightarrow{(d_i)_p} (\mathcal{F}_{i+1})_p \longrightarrow \cdots$$

is exact as a sequence of abelian groups.

Proof. Assume first that the sequence of sheaves is exact. Then for every i,

$$\operatorname{im}(d_{i-1}) = \ker(d_i).$$

Taking stalks at p, we obtain

$$(\operatorname{im} d_{i-1})_p = (\ker d_i)_p.$$

But stalks commute with kernels:

$$(\ker d_i)_p = \ker((d_i)_p),$$

and stalks of sheaf images are images of stalk maps:

$$(\operatorname{im} d_{i-1})_p = \operatorname{im}((d_{i-1})_p).$$

Therefore

$$\operatorname{im}((d_{i-1})_p) = \ker((d_i)_p).$$

Hence the stalk sequence is exact.

Conversely, assume that for every $p \in X$, the stalk sequence is exact. We must show

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

as sheaves.

First, the composition

$$d_i \circ d_{i-1}$$

is zero. Indeed, on stalks,

$$(d_i)_p \circ (d_{i-1})_p = 0$$

for every p. Since morphisms of sheaves are detected on stalks,

$$d_i \circ d_{i-1} = 0.$$

Thus

$$\operatorname{im}(d_{i-1}) \subseteq \ker(d_i).$$

Now let

$$s \in \ker(d_i)(U).$$

Then

$$d_i(s) = 0.$$

For every $p \in U$, the germ s_p satisfies

$$(d_i)_p(s_p) = 0.$$

Therefore

$$s_p \in \ker((d_i)_p).$$

By exactness of the stalk sequence,

$$s_p \in \operatorname{im}((d_{i-1})_p).$$

Hence there exists a germ

$$t_p \in (\mathcal{F}_{i-1})_p$$

such that

$$(d_{i-1})_p(t_p) = s_p.$$

Choose a representative

$$t \in \mathcal{F}_{i-1}(V)$$

on some neighborhood $V \subseteq U$ of p. Then

$$(d_{i-1}(t))_p = s_p.$$

Equality of germs means that, after possibly shrinking V,

$$d_{i-1}(t) = s|_V.$$

Thus s is locally in the image of d_{i-1}.

By the local description of the sheaf image,

$$s \in \mathrm{im}(d_{i-1})(U).$$

Therefore

$$\ker(d_i) \subseteq \mathrm{im}(d_{i-1}).$$

Together with the opposite inclusion, this gives

$$\ker(d_i) = \mathrm{im}(d_{i-1}).$$

Thus the sequence is exact. □

10.1 Exactness versus exactness on global sections

Warning 10.2. *Exactness of sheaves does not necessarily imply exactness on global sections.*

Suppose

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is a short exact sequence of sheaves.

This means that the sequence is exact locally, or equivalently on stalks.

However, the sequence of global sections

$$0 \longrightarrow \mathcal{F}'(X) \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{F}''(X) \longrightarrow 0$$

need not be exact on the right.

The map

$$\mathcal{F}(X) \longrightarrow \mathcal{F}''(X)$$

may fail to be surjective.

The reason is that a global section of $\mathcal{F}''$ may lift locally to sections of $\mathcal{F}$, but the local lifts may fail to glue into one global section.

This failure is measured by sheaf cohomology.

Thus:

$$\boxed{\text{Sheaf exactness is local, whereas exactness of global sections is global.}}$$

10.2 The exponential sequence

A classical example is the exponential sequence on a complex manifold X:

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}_X^\times \longrightarrow 0.$$

This is exact as a sequence of sheaves.

The kernel of the exponential map consists of locally constant integer-valued functions, so it is $\mathbb{Z}$.

The map is locally surjective because every nowhere-zero holomorphic function has a local holomorphic logarithm.

However, it need not be surjective on global sections. On $X = \mathbb{C}^\times$, the function

$$z \mapsto z$$

is a global section of $\mathcal{O}_X^\times$, but it has no global holomorphic logarithm.

Thus this exact sequence of sheaves does not necessarily give an exact sequence of global sections.

10.3 Sheaves form an abelian category

The category Sh_X of sheaves of abelian groups on X is an abelian category.

This means that kernels, cokernels, images, coimages, and exact sequences exist and behave analogously to the category of abelian groups.

However, some constructions require sheafification.

For a morphism

$$f : \mathcal{F} \to \mathcal{G},$$

the kernel is computed objectwise:

$$(\ker f)(U) = \ker\left(\mathcal{F}(U) \to \mathcal{G}(U)\right).$$

But the cokernel is generally computed by first taking the presheaf cokernel and then sheafifying:

$$\mathrm{coker}_{\mathrm{Sh}}(f) = \left(U \longmapsto \mathrm{coker}\left(\mathcal{F}(U) \to \mathcal{G}(U)\right)\right)^+.$$

Similarly, the image is

$$\mathrm{im}_{\mathrm{Sh}}(f) = \left(U \longmapsto \mathrm{im}\left(\mathcal{F}(U) \to \mathcal{G}(U)\right)\right)^+.$$

Therefore:

$$\boxed{\text{Kernels are objectwise, but images and cokernels must be sheafified.}}$$

10.4 Monomorphisms, epimorphisms, and stalks

For sheaves of abelian groups, many categorical properties can be checked on stalks.

Proposition 10.1. *Let*

$$f : \mathcal{F} \to \mathcal{G}$$

be a morphism of sheaves.

Then:

1. *f is injective if and only if every*

$$f_p : \mathcal{F}_p \to \mathcal{G}_p$$

is injective.

2. *f is surjective as a morphism of sheaves if and only if every*

$$f_p : \mathcal{F}_p \to \mathcal{G}_p$$

is surjective.

3. *f is an isomorphism if and only if every*

$$f_p : \mathcal{F}_p \to \mathcal{G}_p$$

is an isomorphism.

Proof. The first statement follows because injectivity is equivalent to the vanishing of the kernel, and kernels are detected on stalks.

The second statement follows because surjectivity of a sheaf morphism means that its image sheaf is the whole target sheaf. This can be checked on stalks:

$$(\operatorname{im} f)_p = \operatorname{im}(f_p).$$

Thus $\operatorname{im} f = \mathcal{G}$ if and only if

$$\operatorname{im}(f_p) = \mathcal{G}_p$$

for every p.

The third statement follows from the first two. □

11 Conceptual summary

The theory of sheafification, sheaf images, and stalkwise exactness is built around the following chain of ideas.

First, a presheaf assigns data to open sets, but it may fail to glue compatible local data.

Second, sheafification fixes this problem:

$$\mathcal{F} \longmapsto \mathcal{F}^+.$$

It preserves all stalks:

$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

It satisfies the universal property:

$$\mathrm{Hom}_{\mathrm{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathrm{PreSh}_X}(\mathcal{F}, i\mathcal{G}).$$

Thus sheafification is left adjoint to the inclusion

$$i : \mathrm{Sh}_X \to \mathrm{PreSh}_X\,.$$

Third, the image of a sheaf morphism is not merely the objectwise image. It is the sheaf of sections locally in the image:

$$(\mathrm{im}\, f)(U) = \{s \in \mathcal{G}(U) \mid s \text{ is locally of the form } f(t)\}.$$

Fourth, exactness of sheaves is local exactness:

$$\mathrm{im}(d_{i-1}) = \ker(d_i)$$

means that every local solution of

$$d_i(s) = 0$$

is locally of the form

$$s = d_{i-1}(t).$$

Finally, because stalks capture local behavior, exactness can be checked on stalks:

$$\boxed{\text{A sequence of sheaves is exact } \iff \text{ the corresponding sequence of stalks is exact at every point.}}$$

This is one of the main reasons stalks are indispensable in sheaf theory, algebraic geometry, complex geometry, differential geometry, and cohomology.